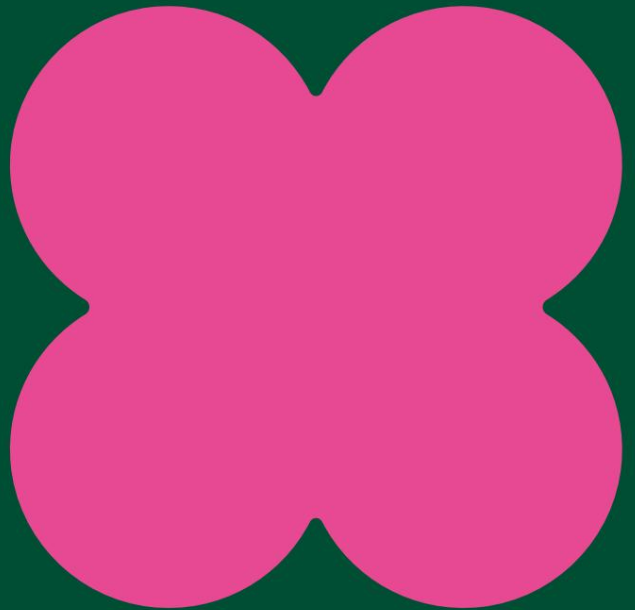
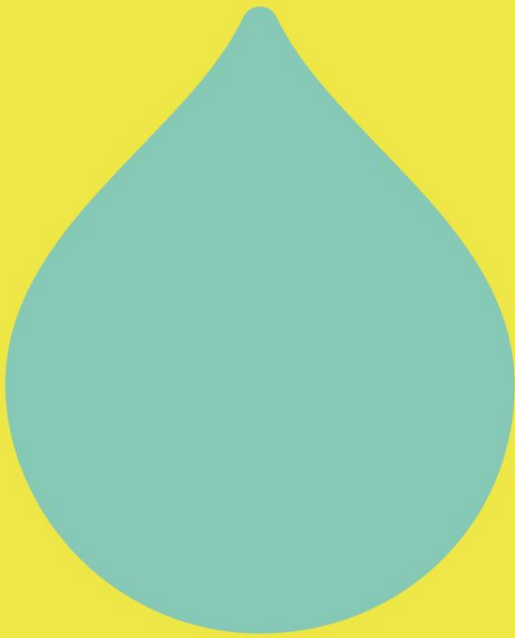
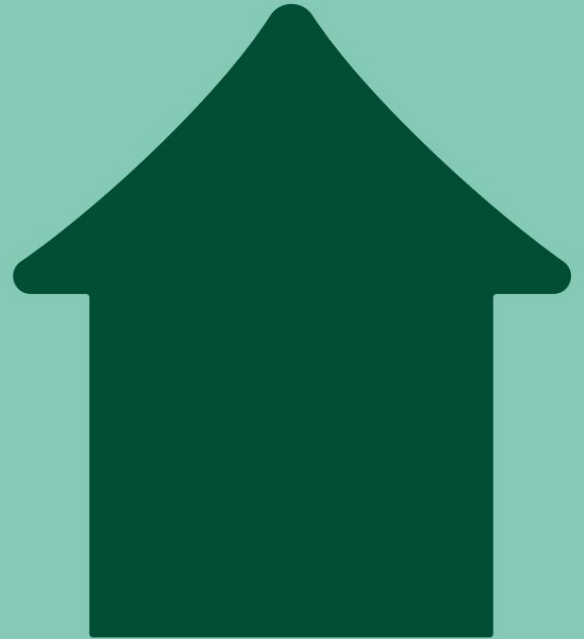
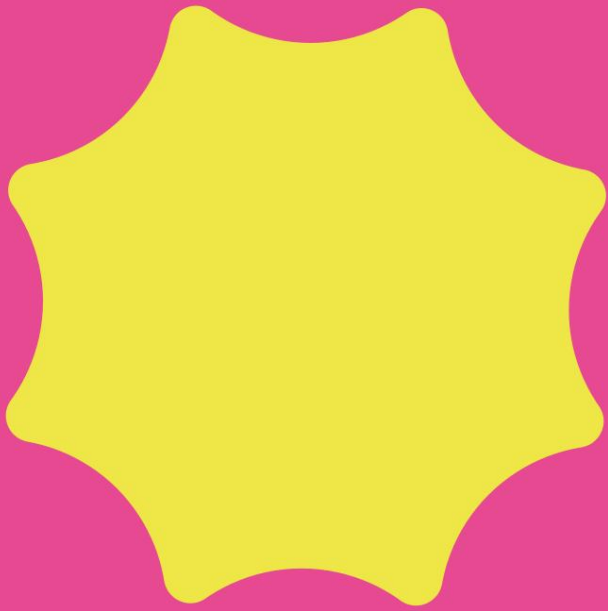




Strategy for coping with the heat.

Heat Plan 2025-2035

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**Ajuntament de
Barcelona**

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Presentation

Barcelona is a city with a long history of commitment to sustainability, social justice, and urban innovation. In the context of the global climate crisis, these values take on a new dimension and urgency. The sustained increase in the planet's average temperature and the intensification of extreme weather events, such as heat waves, are not future scenarios: they are already a reality that directly impacts the health, well-being, and quality of life of its citizens.

The **Heat Strategy: Heat Plan 2025-2035** is the city's strategic response to this challenge. It is an ambitious, rigorous, and comprehensive roadmap that establishes the necessary actions to adapt Barcelona to a scenario of more frequent, intense, and persistent heat. We are doing this with institutional responsibility, but also with a focus on the people. Because we know that **heat does not affect everyone equally** and that its effects can exacerbate pre-existing social, territorial, and health inequalities.

This plan stems from the conviction that decisive and coordinated action is needed to protect the lives and well-being of people, especially those in vulnerable situations: the elderly, children, people with chronic health conditions, people experiencing energy poverty, people experiencing homelessness, and people with reduced mobility. Today, guaranteeing the right to a livable and healthy city is a matter of **climate and social justice**.

The **Strategy to Tackle Heat**. The **Heat Plan 2025-2035** establishes seven major objectives that address the complexity of the heat challenge from multiple dimensions: the transformation of public spaces and urban infrastructure to make them more resilient; the improvement of thermal comfort in homes and public facilities; the strengthening of emergency protocols; the protection of at-risk groups; the promotion of a new heat culture that includes awareness and citizen participation; applied research and innovation; and the adaptation of municipal public policies with a cross-cutting climate perspective.

This Plan falls within the **Climate Plan** and aligns with other strategic initiatives such as the **Natura Plan**, the **Neighborhood Plan**, and the **School Climate Plan**. Among the most noteworthy actions are **the expansion of the climate shelter network**—which already offers nearly 400 comfortable spaces throughout the city—, the strengthening of services for the elderly and dependent individuals, the installation of climate control in public schools, the transformation of school playgrounds, and the promotion of urban shade and green infrastructure.

This document is the result of collaboration between various municipal departments, local organizations, experts, and organized citizens. It is, therefore, an example of how public policies gain strength and effectiveness when built on cooperation, participation, and active listening.

I want to express my sincere gratitude to all the people and institutions that contributed to its development. With this Plan, **we are taking another step towards a more livable, healthy Barcelona, better prepared for climate challenges**. It demonstrates our commitment as a city to protecting the present and ensuring a better future for everyone.

I invite everyone—citizens, professionals, organizations, and social actors—to join us in this great collective challenge. Because tackling the heat is also an opportunity to rethink and improve the city through well-being, sustainability, and care.

Jaume Collboni

The Mayor of Barcelona



Strategy for coping with the heat. Heat Plan 2025-2035

Executive Summary



The **Strategy to Tackle Heat**. The **Heat Plan 2025-2035** is part of the strategic framework of Barcelona's **Climate Plan** and responds to the growing evidence of the impacts of climate change on the city, especially regarding rising temperatures and the intensification of heat waves. This document defines the lines of action to make Barcelona a **resilient, adapted, and** heat-protective city, focusing particularly on **the health and well-being of vulnerable citizens**.

Plan Development Process

The Plan has been developed through a **participatory process** inspired by **international experiences**.

The development of the Strategy to address the heatwave. The Heatwave Plan 2025-2035 was based on a comprehensive and participatory methodology that involved both municipal departments and a broad range of social stakeholders. A cross-departmental working group (the steering committee) was established, led by the Office of Climate Change and Sustainability, with representatives from health, emergency services, social rights, districts, and other key areas. In parallel, more than 180 people participated in a deliberative process that included thematic sessions and contributions through decidim.barcelona, as well as reflections from the European project LOCALISED, which focused on climate justice and the voices of people in vulnerable situations.

Several international experiences have also been analyzed to identify inspiring actions adaptable to the Barcelona context. Cities such as Melbourne, Sydney, Medellín, Boston, Paris, Vancouver, London, and Athens have served as references for their commitment to green infrastructure, data-driven territorial planning, action evaluation systems and neighborhood-level vulnerability analysis, participatory methodologies, and heatwave response protocols. In many cases, their strategies have helped to enrich and refine the actions defined in Barcelona's plan, both in terms of objectives and the design of specific initiatives.

Why the Heat Plan? The impacts of rising temperatures

The Strategy to Tackle Heat. The 2025-2035 Heat Plan was created to respond to the sustained increase in global temperatures, which have already exceeded +1.5°C compared to pre-industrial levels. In Barcelona, this phenomenon is exacerbated by factors such as the urban heat island effect, population density, an aging population, and energy poverty.

Projections indicate increases of up to +5.5°C in summer by the end of the century under passive scenarios. The heat negatively impacts various areas, including:

- Public health: Heat is the leading cause of climate-related deaths. In Barcelona, the heat has caused hundreds of deaths, especially among women and people over 75. Not only extreme heat, but also moderate heat, causes illness and death.
- Well-being and social cohesion: domestic comfort is reduced, social isolation increases, especially among vulnerable groups, and this affects mental health and quality of life.
- Infrastructure and services: heat can cause supply disruptions, damage to asphalt, railway networks and energy and communications infrastructure, among others.
- The economy and tourism: there may be a decrease in labor productivity, especially abroad, an increase in public health costs and losses resulting from energy failures.

- The environment: damage to urban vegetation, reduction of biodiversity, increase in pests and health risks (vector-borne diseases) and increase in air pollution.

This warming affects the population **unevenly** depending on **age, gender, socioeconomic status and territory**.

Where are we? Diagnosing the current situation

Barcelona has already implemented several initiatives to mitigate the effects of the heat. Some examples are:

- The Climate Shelter Network: nearly 400 adapted public spaces and facilities.
- Energy Advice Points (PAE): guidance on energy efficiency and rights.
- The municipal heat wave plan: protocol activated in the event of extreme heat.
- The transformation of school playgrounds and the School Climate Plan, which aims to provide climate control to 170 schools by 2029.
- Urban greening and shade projects: more trees, green roofs and shade structures in public spaces.
- Training for caregivers and strengthening of services for the elderly.

These actions reflect that we are not starting from scratch and constitute a basis for the strategy towards 2035.

Where do we want to go? Plan Objectives

This Plan is defined with the mission of making Barcelona a resilient, adapted and protective city against the heat, especially for vulnerable groups, and is based on seven major objectives:

1. **Reduction of the heat island effect** and improvement of the resilience of public space.
2. **Improvement of thermal comfort in buildings**, especially in homes and public facilities.
3. **Protection of vulnerable groups**, such as the elderly, children, people with illnesses or homeless people.
4. **Improvement of emergency protocols** in the event of intense heat waves.
5. **Shared climate culture**, based on communication, awareness and citizen participation.
6. **Promotion of research and innovation** related to heat.
7. **Adaptation of municipal policies** to integrate the climate perspective.

How will we do it? Key lines of action and measures

This plan is structured around several lines of action for all the defined objectives and has a **cross-cutting monitoring and governance** system that involves various municipal departments and citizen participation. It includes an overall budget and an implementation timeline until 2035.



Strategy for coping with the heat. Heat Plan 2025-2035

Plan Development Process



Defining a heat strategy for the city of Barcelona requires a comprehensive approach and the involvement of all stakeholders who, in one way or another, may be affected by heat or have a responsibility to take action. This ensures, on the one hand, a better understanding of the needs, challenges, and opportunities, and on the other hand, empowers all actors involved in shared responsibility and effective action.

For this reason, to develop the Heat Plan, a steering committee was first formed with representatives from all relevant municipal departments, followed by a participatory and deliberative process with various stakeholders in the city. A comparative analysis was also conducted to draw inspiration from similar initiatives in other cities.

The driving team

The driving force behind this initiative includes members from areas involved in urban heat risk management, such as Environment, Health, Emergency Management, and Urban Planning. The creation of this team enables efficient interdepartmental coordination, ensures a holistic and coherent perspective, and facilitates a more effective emergency response. Led by the Office of Climate Change and Sustainability, the team comprises the following bodies:

- Office of Climate Change and Sustainability, representing Services Urban Planning and Public Space Maintenance
- Social Rights Management
- Barcelona Public Health Agency.
- Barcelona Social Emergency and Urgent Care Center (CUESB)
- Civil Protection
- Resilience Department
- Urban Strategy
- Neighborhood Plan
- Communication and participation
- Resource Management
- Sant Andreu district, representing all the districts of the city

The participatory process

Regarding the **participatory process**, beyond the co-production processes that are already permanently and stably established in the city through the Citizen Council for Sustainability for a more sustainable Barcelona, those organizations that may be more interested in the issues of heat have been identified and seven sessions have been organized, in which more than 180 attendees, representatives of different organizations, have participated.

The sessions have aimed to promote debate and the exchange of arguments between citizens or between citizens and municipal officials, in order to gather their opinions and proposals regarding the Plan, with the following objectives:

- **Report** on the approach and criteria for drafting the Heat Plan.
- **To disseminate the technical work** related to the drafting of the Plan.
- **Gather citizen proposals and considerations** through constructive debate to improve their applicability.

Figure 1. Phases of the participatory process.

Source: Own elaboration



Table 1. List of participants in each of the participation sessions.

Source: Own elaboration

Public space	Research and universities		Dwelling
AVV Besòs and Maresme AVV Barceloneta AVV Besòs	300 k ICTA-UAB	IREC.	Barcelona Energy Agency
AVV Left of the Eixample	ASPB	ISGlobal	Association of Promoters of Catalonia
AVV Gracia Nord – Vallcarca	BSC	LATTS	Barcelona Chamber of Urban Property
AVV Joan Maragall del Guinardó	CIMNE	UB	Advisory Council for Older Persons (CAGG)
AVV Paseo de Sant Joan AVV Brings	Ecoservices	WEEK	Engineering Without Borders
AVV Provençals La Verneda	Metropolis Institute-	UOC	Federation of Neighborhood Associations of Barcelona
Council of Associations of Barcelona (CAB)	IERMB	UPC	
Stop Tanatori Platform			

XAPSL	Health and aging	Sustainability and cooperation	
Saint Peter Claver	ACAF	Atta	FAVB
Ceiling	AVV Gran Via, Espronceda, Peru	OUT OF	Claror Foundation
	Caritas Barcelona	Barceloneta Alert	GIXSU
	Can Peguera House	BATEC	GRAIN
	COFB	Font d'en Fargues Village	IE Tres Fonts de les Courts
	People's Advisory Council Older adults (CAGG)	CCOO of Catalonia	Permaculture
	CUS	COEAC	PIMEC
	EAP Sant Rafael	Red Cross	Catalan Society for Environmental Education
	FAVB	Bethany Patmos School	UPC
	"la Caixa" Foundation	ENVIRONMENTAL SPACE	
	Catalan Institute of Health		

Table 2. Sessions, number of participants and number of contributions from the participatory process.

Source: Own elaboration

In addition to the organized sessions, the municipal participatory digital platform decidim.barcelona has been opened, where citizens have been able to incorporate more proposals. The total number of proposals received is summarized in the following table:

Date	Session	Participants' Contributions	
04/02/2025	Inaugural session of the Monitoring Committee (Sustainability Council)	-	-
11/02/2025	Session with research centers and universities	26	7
11/02/2025	Session on public space	22	5
19/02/2025	Session on housing	9	5
04/03/2025	Session with XAPSLL (homeless people)	3	10
05/03/2025	Session on health and aging	21	16
06/03/2025	Session with the Council for Sustainability, the Network B+S and the Cooperation Council	40	16
13/03/2025	Session on municipal facilities	45	9
	Let's decide Barcelona	15	26
TOTAL		181 people, 94 proposals	

The majority of the proposals submitted (76%) have been incorporated into the Plan or are part of other action plans, while a minority (13%) have been discarded due to technical limitations or because they fall outside the scope of the Plan. Some proposals (11%) are currently undergoing technical evaluation as of the date of publication of this document. Feedback has been provided on each contribution.

In addition to the measures incorporated as a result of the participation process, the results and contributions of the project **“Nobody behind: active listening to vulnerable groups to guarantee climate justice”** —included within the framework of the European project **LOCALISED**— **have also been taken into account in the preparation of this document**. The project aims to understand the perspectives and narratives of citizens, especially those from vulnerable groups. As part of this initiative, active listening sessions were designed to share, discuss, and exchange ideas with diverse groups particularly vulnerable to climate change, including children, the elderly, people with disabilities, and people from diverse cultural backgrounds.

Some of these sessions focused on home climate control, energy efficiency and renewable energy, the role of public spaces in preparing the city for climate change, and the use of city spaces and facilities as climate shelters.

Inspiring experiences from other cities

The climate adaptation and heat management plans of numerous cities worldwide have been analyzed to identify actions that can be implemented in Barcelona. Meetings and discussions have been held with many of these cities over the past few years to exchange experiences, either through city networks such as C40 (with the Cool Cities Network) or bilaterally. Some interesting initiatives are summarized below (see Annex I for more information).

In the Australian city of **Melbourne**, the main course of action to combat the heat is related to green spaces and nature. In addition to having a strategy

Adaptation to climate change, mainly related to the decarbonization of the city and the reduction of the impact of waste, has two other action plans: the nature in the city strategy and the urban forest strategy.

Both organizations approach the topic of green spaces from different perspectives. They conduct thorough analyses of the benefits of green spaces in numerous areas and develop strategies to increase their presence in the city. This city also has a heatwave action plan for emergency and social services, which has served as a source of inspiration when compiling a list of vulnerable groups.

A heat management plan developed in the **Sydney** metropolitan area of Australia has also served as inspiration for the plan developed for the city of Barcelona. It details various aspects of interest, from a comprehensive analysis of the effects of heat on people, infrastructure, and society, to the vision and objectives of the actions described.

In Colombia, the city of **Medellín** has received international recognition for its network of green corridors, which has been under development since 2016, further strengthening the role of green spaces as a strategy for addressing the climate crisis. Another inspiring example is the city of **Boston** in the United States. A strategic plan has been developed to tackle urban heat, including a comprehensive neighborhood-by-neighborhood analysis of heat distribution and vulnerability, as well as other relevant indicators. This plan is also notable for its participatory process, giving citizens a significant voice in decision-making. Furthermore, a series of objectives are defined, and each of the outlined actions is evaluated based on its progress toward or away from achieving them, which has been adapted to the specific actions described in the plan.

New York City's Cool Neighborhoods NYC plan combines environmental, social, and monitoring actions to mitigate, monitor, and adapt to the effects of urban heat, with the goal of protecting public health and reducing inequalities in the face of climate change.

Another city that has made significant progress in adapting to climate change and heat is **Paris**, France. In addition to conducting a 50°C heat simulation, which has served as inspiration for our city and shaped one of the action lines included in this plan, they have an integrated heat management plan within their overall climate change adaptation plan. This plan defines 396 actions with various objectives, ranging from decarbonizing the city to greening. Each of these heat-related actions has been compared with the actions designed in this Heat Management Plan. It has been observed that Paris's measures are more specifically designed, resulting in several actions being grouped under a single action line in Barcelona. In some cases, some of the actions described in Paris have helped to enrich the design and description of the actions in this Plan.

The Canadian city of **Vancouver** has taken an innovative approach, designing an action plan for social and emergency services in the event of heat waves based on past experiences. **Madrid**'s Action Plan for High Temperature Episodes aims to reduce morbidity and mortality from heat waves through monitoring, institutional coordination, and preventative measures across various sectors. Meanwhile, in the United Kingdom, the city of **London** presents an interesting contrast between the risks and opportunities arising from climate change.

Finally, one of the cities that has received special attention is **Athens**, since, although it has a considerably smaller population than Barcelona, it shares many other similar characteristics. In addition to having a very similar climate, its population density is of the same order as Barcelona's (approximately 17,000 inhabitants/km²).

km²). Its strategic plan has very similar foundations to Barcelona's. First, it identifies heat as the main risk of climate change and justifies in detail the

need to adapt to the new hot summers to protect the public.

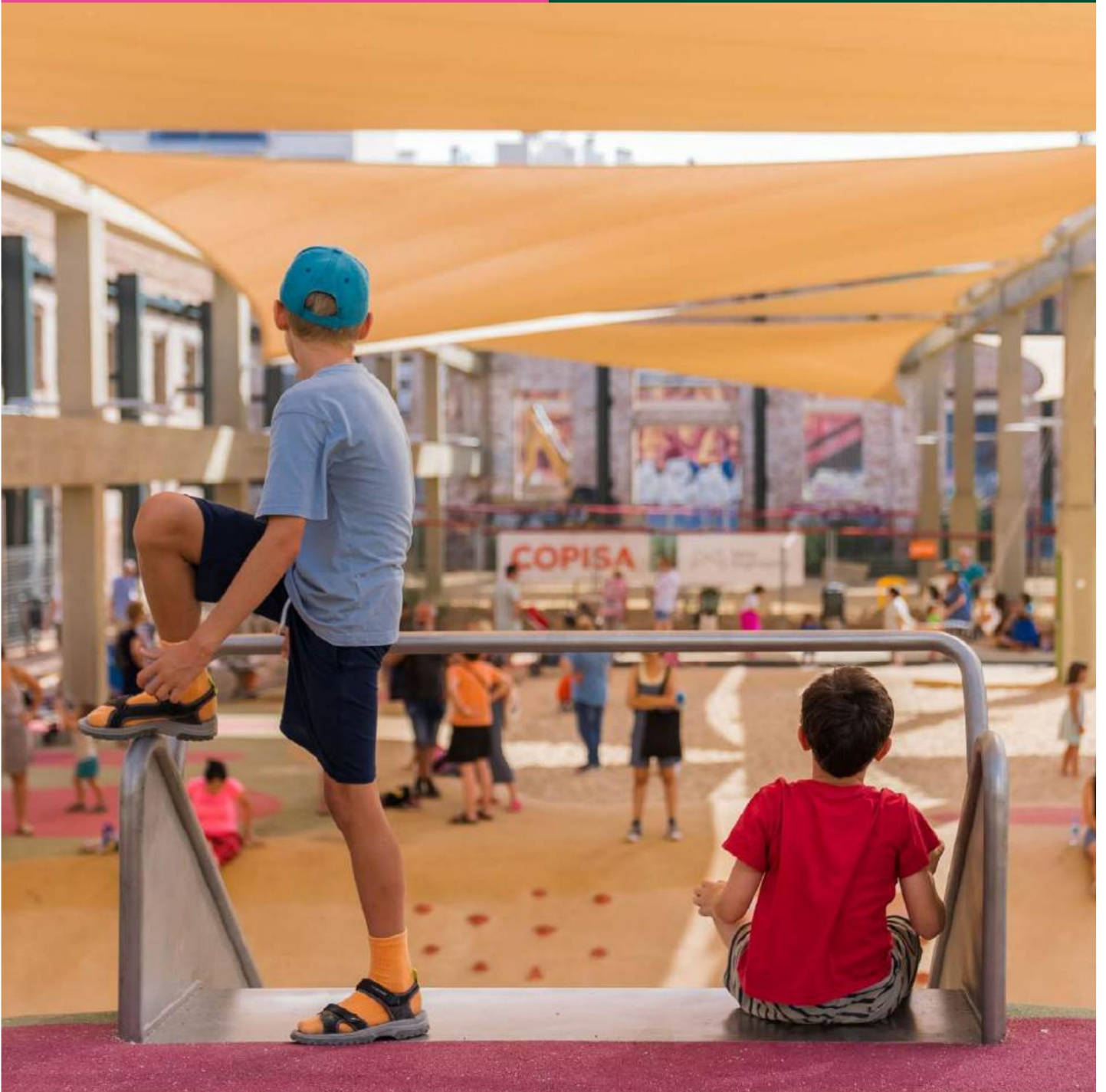
The distribution of heat in the city is studied and compared with the distribution in the most vulnerable sectors. The actions that constitute its strategy are very similar to those described in this Plan.

These cities have been among the most relevant and prominent examples used as inspiration in the creation of this Plan. Consideration has been given not only to the specific solutions implemented in these urban centers, but also to how each has addressed the challenges arising from global warming and extreme heat in innovative ways, adapted to its particular characteristics. Technical contacts have been established with many of them through the C40 network (via the Cool Cities Network) or bilaterally to share experiences and knowledge. It is worth mentioning that, in addition to these cities, the actions and strategies taken by other major cities worldwide, such as Singapore, Washington D.C., Phoenix, and Tokyo, among others, have also been consulted. This process of consultation and study of international experiences has been fundamental in enriching the Plan and ensuring that global best practices are taken into account, with the aim of adopting efficient solutions tailored to local realities, but with a global perspective on climate change.

**Strategy for
coping with
the heat.**

**Heat Plan
2025-2035**

**Why a heat
plan?**



The global temperature has already exceeded the 1.5°C limit set in the 2016 Paris Agreement. This has led to more heat waves, longer summers, and shorter winters. Climate change exacerbates the urban heat island effect, as it makes high temperatures have a greater impact.

Europe is warming twice as fast as the global average, and Barcelona is particularly vulnerable due to factors such as climate, population density, the aging of its citizens, and energy poverty.

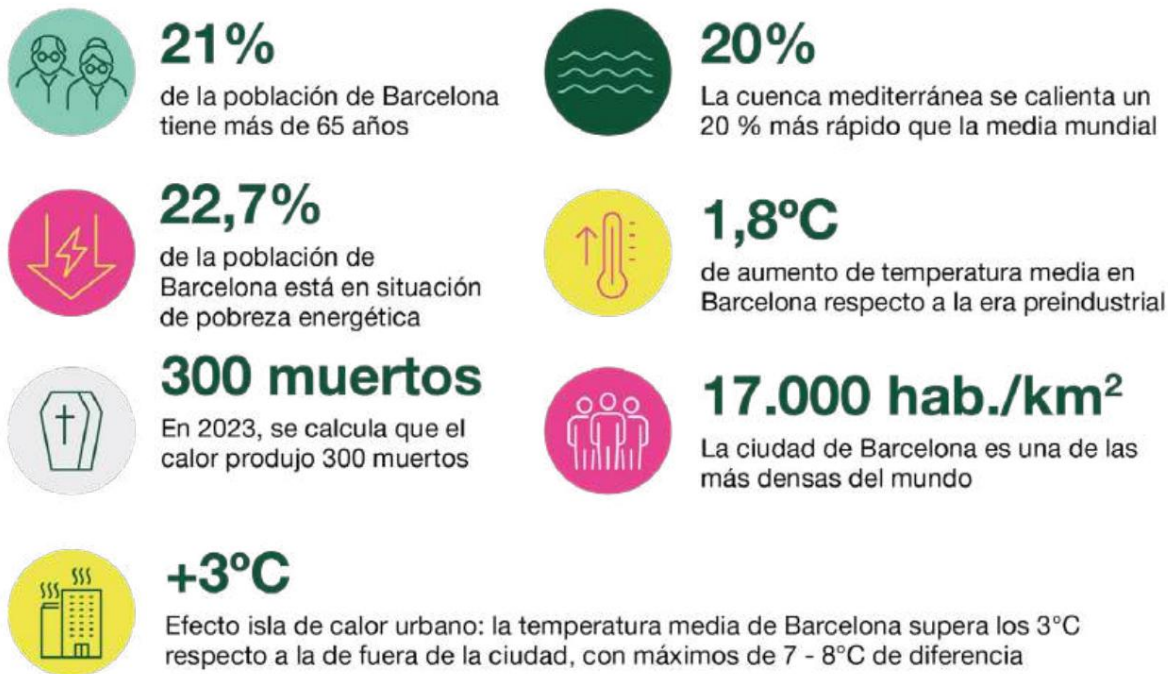
Climate projections indicate that the average temperature in Barcelona will increase significantly. Furthermore, observations confirm this trend.

The effects of climate change in the city are diverse and unequal, and they disproportionately affect the most vulnerable groups. Heat can severely impact health and even cause death, affecting quality of life, but it can also have negative effects on supplies, the cost of living, the environment, and infrastructure, as well as increase the risk of wildfires.

Figure 2.
Barcelona, a
vulnerable city
to the heat.

Source: Prepared by
own

Barcelona, una ciudad vulnerable al calor



The global temperature increase has exceeded 1.5°C

According to data from the Copernicus Programme's Global Climate Highlights 2024 report (1), 2024 saw unprecedented global temperatures, following the remarkable heat of 2023. In fact, the last ten years have been the ten warmest on record. It was also the year in which the average temperature clearly exceeded 1.5°C above pre-industrial levels, the threshold set by the **Paris Agreement**.

of 2016 to significantly reduce the risks and impacts of climate change. One or two years exceeding 1.5°C above pre-industrial levels do not imply a violation of the Paris Agreement. However, with the current rate of warming of

With global warming exceeding 0.2°C per decade, the likelihood of surpassing the Agreement's 1.5°C target by 2030 is highly probable. Some projections indicate that, if current emissions rates continue, the planet's average temperature could rise by more than 3°C by the end of this century.

These temperatures in Europe are rising at a rate more than double the global average. While the global temperature increase has been 0.2°C per decade, in Europe it has been 0.5°C.

Barcelona is a city especially vulnerable to heat for several reasons.

According to the latest MedECC report(2), the Mediterranean basin is warming 20% faster than the global average and, along with Antarctica, is one of the world's major climate change hotspots. Another study(3) indicates that the **Mediterranean climate** makes cities like Barcelona more acutely affected, as its hot, dry summers are exacerbated by climate change, resulting in more frequent and intense heat waves. In addition to this Mediterranean factor, other factors make Barcelona particularly vulnerable to heat: it is one of the most densely **populated** cities in Europe, has an **aging population** (21% of residents are over 65, according to IDESCAT data), and also has a high percentage of people living in **energy poverty** (22.7% of the population). The study also points out its limitations to adaptation, not only because of its vegetation cover—which is limited, especially in certain more vulnerable areas—, but also because of its geographical boundaries, which prevent the development of measures in the remodeling of the city.

Climate change affects cities in a variety of ways. Some of its effects are tangible, such as those resulting from an episode of intense storms with uprooted trees and flooding. In contrast, the effects of heat are usually easier to underestimate because their impacts often go unnoticed or occur a few days later.

Heat does not affect everyone equally. Factors such as age, health, gender, physical condition, socioeconomic status, and the conditions of the environment where we live and work determine the potential degree of heat-related impact. Therefore, some groups are more vulnerable to its effects, as they have less capacity to anticipate risks and recover from damage, and thus suffer the effects to a greater degree.

The impacts of heat also vary by city district, as they are affected by factors such as geography, the condition of buildings, the presence and type of vegetation, the percentage of impermeable surfaces, urban density, and the dispersion of heat sources.

The climate emergency demands swift action to avert the worst-case scenarios projected. **Alongside accelerating mitigation efforts, adaptation measures are both necessary and urgent** to reduce the impact of these phenomena. In this regard, several international initiatives should be followed, including:

- The guide Heatwaves and Health: Guidance on Warning-System Development, developed by the World Health Organization (WHO) and the World Meteorological Organization (WMO), describes different intervention strategies to address heat waves and their effects, as well as providing guidance on heat warning systems. It also recommends that countries develop and implement heat response plans.

In Catalonia, there is an operational plan to prevent the effects of heat on health (POCS).

The Global Cooling Pledge, launched at the 2023 United Nations Climate Change Conference (COP28) and signed by several states, including Spain, aims to reduce cooling-related emissions by 68% compared to current levels by 2050, significantly increase access to sustainable cooling by 2030, and increase the global average efficiency of new air conditioners by 50%. This commitment recognizes the fundamental role of cities in combating urban heat and recommends developing heat action plans by 2026, increasing green spaces and improving thermal comfort in a sustainable and efficient manner.

– The UN Secretary-General's Call to Action on Extreme Heat, which calls for an urgent and concerted effort to strengthen international cooperation to address extreme heat in four key areas: caring for vulnerable people, protecting working people, strengthening the resilience of economies and societies through data and science, and limiting the temperature increase to 1.5°C by phasing out fossil fuels and increasing investment in renewable energy.

In line with what other major cities around the world are doing, and with the aim of reducing the effects of heat, **Barcelona has determined the urgency of taking the necessary measures to prepare the city for high temperatures and heat waves during the summers.**

It's very hot in Barcelona.

Climate projections in Barcelona

According to climate projections made in Barcelona by the Meteorological Service of Catalonia for various scenarios (see the Climate Emergency Action Plan 2030), an increase in average temperatures of 1.7 °C is expected by the end of the century for the committed scenario RCP4.5 and an increase of up to 3 °C by the end of the century for the passive scenario RCP8.5.

Regarding heat waves, in the period 1982-2015 there was one heat wave every four years, and it is expected that by the end of the century there will be two heat waves every year in a committed scenario (RCP4.5) and four to five annual heat waves in a passive scenario (RCP8.5).

According to climate forecasts for the end of the century, other indicators related to extreme temperatures are also expected to increase, such as hot days (maximum temperature >30 °C), torrid days (maximum temperature >35 °C), tropical nights (minimum temperature >20 °C) and torrid nights (minimum temperature >25 °C)(4).

It is worth noting that the increase in average temperatures will be much more pronounced in summer. According to the results of the RESCCUE project, by the end of the century, average summer temperatures could increase by 3°C in the compromise scenario and by 5.5°C (with a maximum of 8.5°C) in the passive scenario. Furthermore, warmer winter temperatures are also expected, rising by 1.5°C and 3°C by the end of the century for the passive and compromise scenarios, respectively. This will result in much milder winters than those currently experienced.

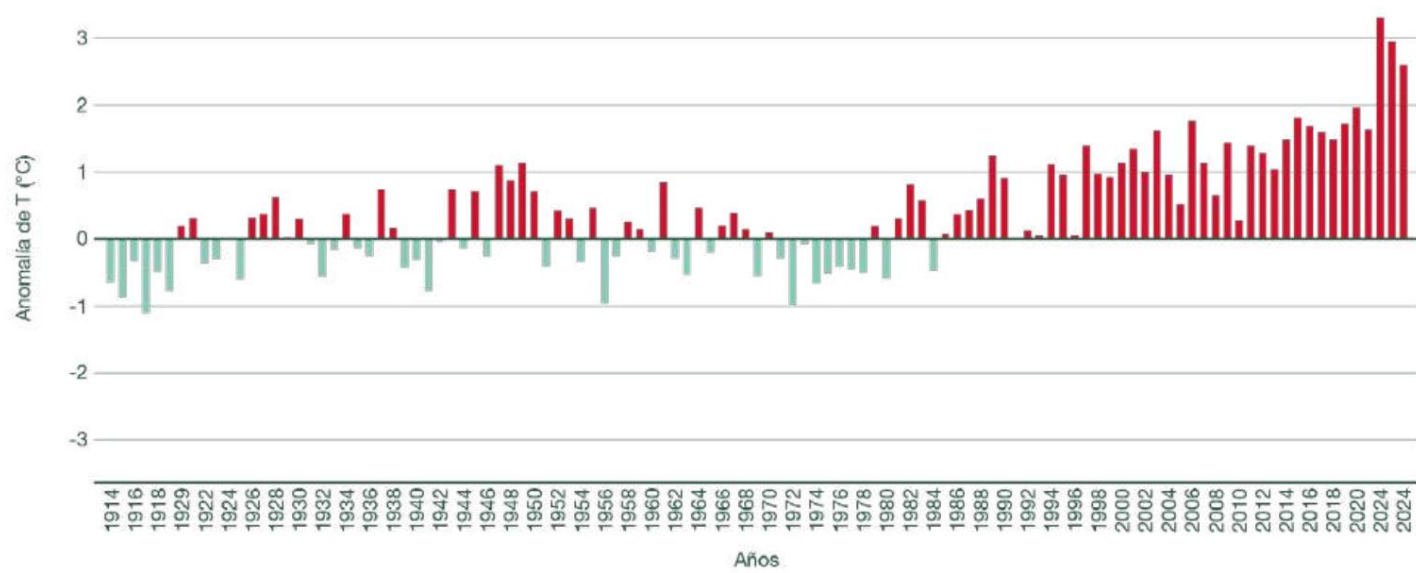
The actual data confirms the trend towards increasing temperatures

Based on the observation of historical temperature data, the trend of increasing temperatures from the end of the 18th century to the present is confirmed in Barcelona.

The evolution of the average annual temperature shows a clear upward trend from the beginning of the 1990s to the present. The increase since the pre-industrial era has in some cases been as much as 1.8 °C (2022).

Figure 3. Annual mean temperature anomaly (1783-2024). The graph shows the annual mean temperature, but expressed as an anomaly with respect to the reference period 1961-1990. The blue bars indicate years with a negative anomaly (or cold years) and the red bars show years with a positive anomaly (or warm years).

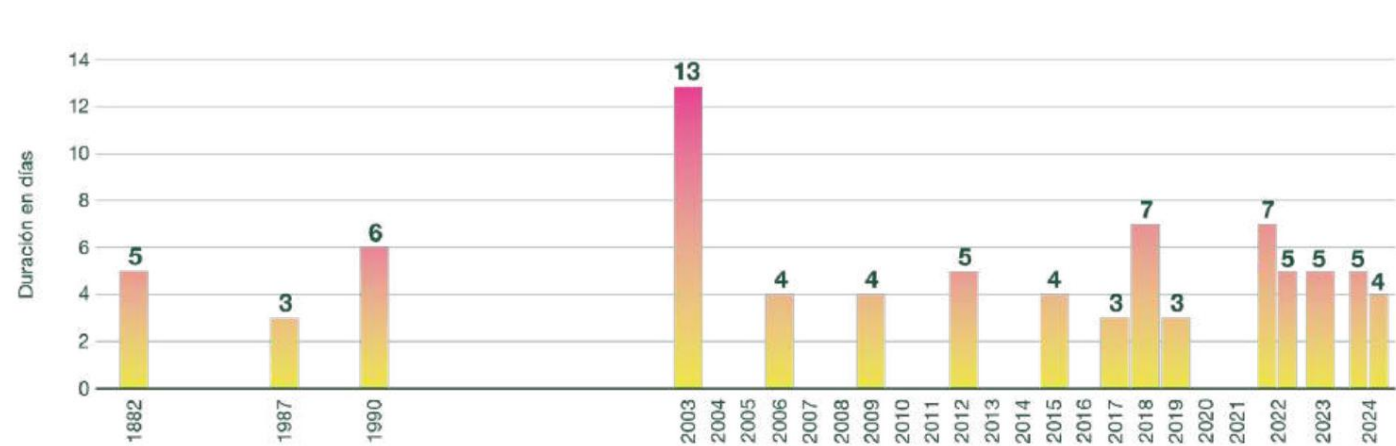
Source: Meteorological Service of Catalonia



Regarding **heat waves**, an increase has been detected in the number of heat waves experienced in Barcelona during the summers, as well as an increase in the number of days these waves last. Between 1982 and 2015, it was estimated that Barcelona experienced one heat wave every four years. In 2022, we experienced two, one in 2023, and two in 2024 (Figure 4).

Figure 4. Evolution of the duration of the different heat waves that Barcelona has had in the period 1982-2023.

Source: Barcelona Regional, based on data from various weather stations, 2024.



A heat wave occurs when a given area experiences three consecutive days of extreme temperatures. In Catalonia, this threshold is the 98th percentile of maximum temperatures.

Regarding hot days and nights (tropical and torrid days and nights), with the temperature data we have available for the city, it can already be seen how, in recent years, the number has been increasing.

Table 3. Evolution of heat waves, hot and scorching days, and tropical and scorching nights from the period 1982–2015 to the years 2022, 2023, and 2024, and a comparison with respect to the compromised and passive scenarios for the end of the century. The data for 2023 and 2024 correspond to the average of the meteorological stations in Barcelona, excluding the Fabra Observatory.

Source: Prepared by the author using data from the Meteorological Service of Catalonia

Indicator	Starting data resulting from the average of the period 1982-2015	2020 Data	2021 Data	2022 Data	2023 Data	Data 2024	Projections scenario, committed, end of the century	Projections, passive, century
 Heat wave/year (the maximum temperature exceeds the 98th percentile of the three hottest months of the year). For 2020 This threshold was 33.6 °C	0,25 1 every 4 years	0	0	2	1	2	2	4-5.
 Warm day (>30 °C)/year	22	37 26 64 42 52					50	80
 Hot day (>35 °C)/year	0,5 1 every 2 years	0	0	2	4	3	2,5	8,5
 Tropical night (>20 °C)/year	38	74 88 105 82 86					83	112
 Torrid night (>25 °C)/year	1	7	1	14	13 23		2,5	6
 Hellish nights (>30 °C)/year	sin data	sin data	sin data	0 0		0	sin data	sin data

The urban heat island effect is a factor to consider

The urban heat island effect is a phenomenon that occurs in cities, where temperatures in urban areas are significantly higher than in the surrounding rural areas. This effect is due to the fact that materials present in cities, such as asphalt and concrete, absorb and retain heat during the day and release it slowly at night, so the temperature does not drop as it would in natural areas. Factors such as the reduction of vegetation in cities, as well as increased mobility, exacerbate this effect.

The greatest intensity of the urban heat island effect in Barcelona occurs at night and during the winter. Urban weather stations record temperatures up to 3°C higher (annual average) than those outside the city, with maximum observed temperature differences of up to 7 or 8°C.

Climate change intensifies the urban heat island effect, as it increases exposure to episodes of high temperatures and heat waves, thus exacerbating their consequences.

The nighttime urban heat island effect intensifies the effects produced by the heat, especially all those related to health, since the nighttime rest period is the time when people recover from the heat suffered throughout the day.

According to a recent study(3), urban heat islands increase mortality by 4% during the summer period.

This highlights the need to develop lines of action whose main objective is to reduce the urban heat island effect.

Thermal comfort depends on factors other than temperature.

When evaluating the effectiveness of environmental measures implemented to combat heat, one of the indicators to consider is **thermal comfort**. The ISO 7730:2005 standard on thermal environment ergonomics defines thermal comfort as "**the thermal sensation experienced by a human being in relation to the overall thermal equilibrium of the body**," that is, the feeling of comfort a person can experience in relation to the outside temperature based on their internal temperature. It is a subjective measure, but there are still several methodologies for evaluating it.

To monitor thermal comfort, parameters that can be considered, in addition to temperature, include humidity, wind speed, and solar radiation. When we talk about temperature, we're not just referring to air temperature, but also to radiant temperature, which is the temperature of surfaces surrounding a person, such as the floor, walls, and buildings.

Thermal comfort can be assessed using simulation models that integrate many of these parameters. Some of the most commonly used models in Barcelona are the Universal Thermal Climate Index (UTCI), the Wet Bulb Globe Temperature (WBGT), and the Heat Index. The model used depends on the characteristics of the location (outdoors, indoors, sunny, shaded, etc.) and the purpose of the measurement itself.

Other institutions, such as The American Society of Heating, Refrigerating and Air-Conditioning Engineers (ASHRAE), incorporate personal factors into the concept of thermal comfort, such as metabolism (physical activity) or thermal insulation (according to the type of clothing).

Figure 5.

Environmental
and personal
factors related to thermal
comfort.

Fuente: United
Nations Environment
Programme (2021).
Beating the Heat: En
Sustainable Cooling
Handbook for Cities.
Nairobi.

Factores ambientales relacionados con el confort térmico



Temperatura radiante

Calor que irradia un objeto caliente, tales como hornos, superficies calientes y maquinaria



Temperatura del aire

Temperatura del aire que rodea el cuerpo



Velocidad del viento

Velocidad del aire en movimiento. El aire que se mueve en condiciones de calor o humedad puede incrementar la pérdida de calor del cuerpo a partir de la convección sin ningún cambio en la temperatura, lo que ayuda al enfriamiento



Humedad

Vapor de agua en el aire. La habilidad del aire de retener vapor de agua se relaciona directamente con su temperatura. Cuanto más caliente es el aire, más humedad o vapor de agua puede retener

Factores personales relacionados con el confort térmico



Ropa

Ropa adecuada a las condiciones climáticas



Tasa metabólica

Inherente a la persona y determinada por su nivel de actividad

Heat has significant consequences for health and well-being, public space, and infrastructure.

The most obvious effects of heat are those that directly affect people's **health and well-being**, but they are not the only ones to consider. Sustained exposure to high temperatures also has significant consequences for infrastructure and the daily functioning of the city, compromising the provision of basic services and hindering the response of emergency services.

Furthermore, the heat generates significant **economic losses**, both due to decreased productivity and the costs associated with emergency management and the adaptation of equipment and services. Although quantifying these losses can be complex, they have a real impact that must be taken into account.

Finally, it is also worth highlighting the **effects on the urban natural environment**, since the vegetation, fauna and local ecosystems are seriously altered by the high temperatures, with environmental and social repercussions in the medium and long term.

Figure 6.
Effects of heat.

Source: Own
elaboration

Efectos del calor



About health and well-being

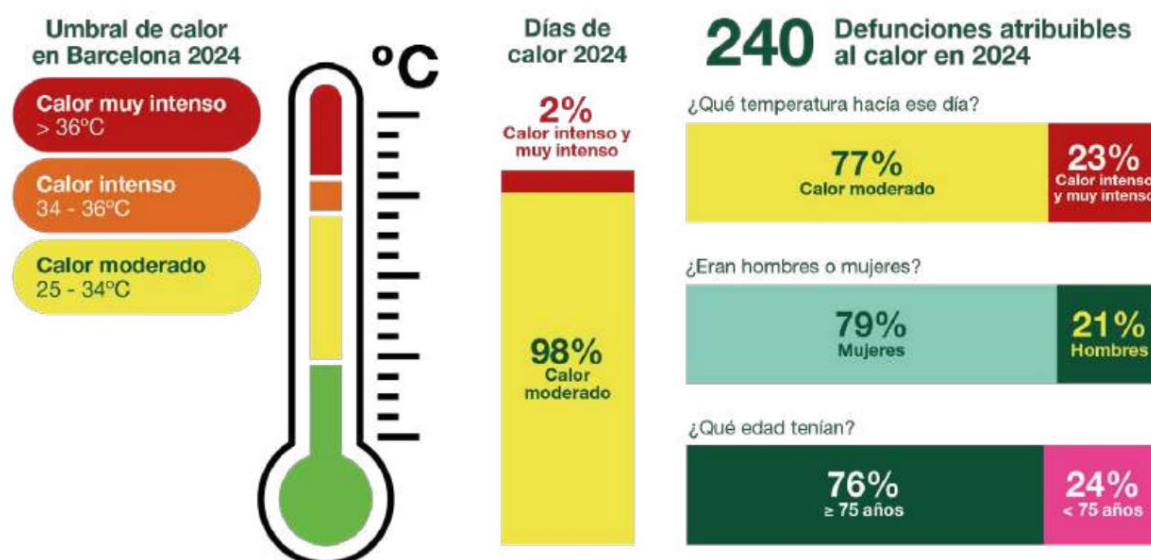
Among the various **effects of heat**, it is worth highlighting those related to **a decrease in people's health and quality of life**, especially in vulnerable population sectors.

Extreme temperatures are not necessary to suffer the effects of heat. Although the more intense the heat, the greater the mortality and morbidity it causes, even moderate heat has significant health effects. According to 2024 records from the Raval weather station, the intense heat threshold, set at 34.1 °C for that year, was exceeded on three days during the activation period of the POCS (Operational Plan to Prevent the Effects of Heat on Health), which runs from May to September. On two of those three days, the very intense heat threshold (36.1 °C) was also exceeded. The remaining days fell within the moderate heat range. According to data from the ASPB (5), approximately 75% of heat-attributable mortality occurred on days of moderate heat. Furthermore, the data clearly indicate that women and people over 75 years of age are more severely affected by heat.

Figure 7.
Relationship between
heat and mortality
in Barcelona.

Source: ASPB Report
"Mortality attributable to
heat in the city of
Barcelona 2016-2023"

Relación entre calor y mortalidad en Barcelona - 2024



The effects of heat on health can be classified as direct, if they result from exposure to heat, or indirect, if they are related to the environmental and social consequences caused by climate change and rising temperatures.

Direct effects

An increase in heat translates into an increase in mortality and morbidity.

The body's ability to regulate internal temperature is compromised, resulting in negative health effects such as heat exhaustion, heat cramps, and heat stroke, which in the most severe cases can lead to death.

According to a study published in the journal *Nature Medicine*(6), it is estimated that in 2003 heat caused 71,449 deaths in Europe during the months of June, July, August, and September. In 2023, during the same period, 61,672 deaths attributable to heat were recorded, with Spain being one of the most affected countries.

In Barcelona, the Barcelona Public Health Agency (ASPB) estimated that between 1992 and 2015, there were 980 deaths in men and 2,729 in women attributable to extreme heat (natural deaths of people aged 25 and over), generally among the elderly and frail individuals (4). More recent data show that in 2024, during the months of May to September, heat-related mortality increased, resulting in approximately 240 deaths, representing 5% of total mortality during those months. These deaths were concentrated primarily among women (around 190 deaths) and people aged 75 and over (around 180 deaths) (5). Furthermore, it has been observed that in Barcelona, nighttime heat is associated with higher mortality from natural, cardiovascular, and respiratory causes. Specifically, when nighttime temperatures exceed 23 °C, the risk of mortality increases significantly (4).

It is important to consider the difficulty in counting heat-related deaths, as it often causes deaths not as the main cause, but **by aggravating pre-existing conditions** in people, such as certain cardiovascular, respiratory, cerebrovascular, gastrointestinal diseases (especially in children), or in the case of diabetes, kidney disorders (including kidney failure), mental, behavioral and cognitive disorders and premature birth.

Heat can aggravate these diseases to the point of eventually causing death. These types of deaths are not directly attributed to the heat, but to the disease in question, which translates into an underestimation of mortality caused by the heat(4).

A relationship has also been found between exposure to intense heat and an **increase in hospital admissions**, emergency room visits, and ambulance transfers.

Likewise, in a more subtle way, an association has been found between heat and the number of calls to emergency medical services for milder matters that do not require any admission, medical visit or ambulance transfer(4).

Heat also affects academic performance and concentration, especially in educational or work environments that are not adequately air-conditioned. High temperatures interfere with cognitive ability and impair attention, memory, and decision-making (7). In schools, this translates into decreased student performance, with particularly noticeable effects during periods of extreme heat or in classrooms that lack adequate cooling systems.

In the workplace, heat can cause fatigue, irritability, and reduced productivity, as well as increase the risk of errors and accidents, especially in manual or physical work (8). This impact not only affects learning and the quality of work but can also exacerbate social inequalities, since people who study or work in poorly ventilated buildings often belong to more vulnerable groups.

However, in recent decades, **studies have observed a decrease in the risk of heat-related mortality during the summer (9), (10)**. This decrease could be related to various factors, such as physiological adaptation; changes in habits, such as wearing cooler clothing or greater awareness of the health risks of heat; structural improvements, such as increased use of air conditioning, better housing conditions, or a more favorable general health status of the population; as well as the implementation of specific plans to reduce the health effects of heat waves.

Indirect effects

Heat can significantly impact daily well-being and **disrupt the quality and habits of daily life**, especially in urban environments like Barcelona. When temperatures are very high, the feeling of comfort decreases drastically, both indoors and outdoors. Sleep suffers (especially without air conditioning), people adopt more sedentary routines, and tend to avoid outdoor activities. Public spaces such as parks, squares, and streets become less livable during the hottest hours of the day, limiting opportunities for socialization, leisure, and active mobility (such as walking or cycling) and reducing the use of community spaces. All of this translates into an increase in cases of social isolation, as people stay home to avoid the effects of extreme conditions.

According to studies conducted in the Australian city of Sydney(9), this **social isolation** This can easily lead to mental health problems, which are often exacerbated by the lack of physical exercise and the inability to go outside.

Rising temperatures also lead to an **increase in the cost of living**, reflected especially in the rise in energy prices—due to increased demand—and food prices, as the heat negatively impacts agricultural production and reduces its availability. This scenario also contributes to increased energy poverty, since many households cannot afford to maintain a comfortable temperature. According to the results of the PENSA project in 2020(11), energy poverty is linked to a poorer perception of health, more fragile mental health, a lower quality of life, and a higher prevalence of chronic diseases.

The increase in temperature can lead to a decrease in the climate comfort of passengers and staff, which translates into a possible **decrease in the use of public transport** in favor of private transport.

On the infrastructure and functioning of the city(12)

High temperatures have **direct impacts on essential supplies and the functioning of urban infrastructure**.

High temperatures and droughts can reduce **water availability**. High temperatures are also expected to lead to increased **energy demand**, due to greater energy consumption for air conditioning and increased electrification of buildings and transportation. This increased demand may lead to a greater risk of power outages due to potential system overload. Furthermore, rising temperatures may affect the electricity generation efficiency of thermal and nuclear power plants that supply Barcelona, as well as cause losses in the electrical grid or overheating of lines or transformers.

Furthermore, extreme heat can cause **physical damage to urban infrastructure**, as has been observed in other cities that have experienced higher temperatures than Barcelona. Asphalt can crack or degrade more rapidly due to the premature aging of its components, and transportation systems, such as rails and overhead lines, can deform due to thermal expansion, increasing the risk of derailments.

Researchers from the Polytechnic University of Madrid (UPM) have analyzed the effects that changes in environmental conditions can have on critical sections of Spanish roads and railways, and conclude that **7% of road sections and 14% of the railway network could be affected in the future**.

The Government of Catalonia carried out the Assessment of the Vulnerability of Catalonia's Transport Infrastructure to the Impacts of Climate Change (Catalan Office for Climate Change and Estudi Ramon Folch & Associats, October 2021). In the case of Barcelona, the impact of heat was primarily felt in the railway infrastructure (train and metro), as shown in the following maps. It should be noted that this assessment requires further analysis at a more detailed scale, incorporating the specific characteristics of each infrastructure.

Figure 8.
Road (left) and rail
(right) network sections
in areas with y1.75 heat
waves in

summer.

Source: Report

Barcelona Regional
March 2025:

Evaluation of the
vulnerability of the
infrastructures of

Transport from Catalonia to
the impacts of climate
change



Heat also affects buildings . It can accelerate the aging and deterioration of building materials such as concrete, steel, and wood, leading to cracks and warping. Furthermore, the expansion and contraction of facade and cladding materials can cause premature damage and reduce their lifespan.

High temperatures are also expected to affect the performance of **electronic components** in electrical cabinets that control signaling, communication, and lighting systems. The resulting power outages could compromise communications and severely complicate the coordination of **emergency services**. This situation could be exacerbated by the impact of direct and persistent heat on vehicles, which may suffer engine overheating or failures in sensitive components such as tires.

About the economy

Heat also has a **negative impact on labor productivity and working conditions**, especially in sectors exposed to the elements. Heat waves, in particular, have a considerable economic impact, especially in sectors such as agriculture, construction, and industry, where productivity decreases due to extreme conditions, occupational health risks increase, and activities can be disrupted. This leads to direct losses from lost work hours and damage to infrastructure and crops, as well as increased energy expenditure for heating and cooling, which can worsen the climate crisis if renewable energy sources are not used.

Sectors such as trade and tourism are also affected, as extreme weather conditions can alter consumption and mobility habits(8).

According to the International Labour Organization (8), it is estimated that by 2030, increased heat stress due to climate change could lead to the loss of 2.2% of total working hours worldwide, equivalent to 80 million full-time jobs. This reduction in productivity would translate into global economic losses of \$2.4 trillion. In fact, between 1980 and 2000, Europe already suffered losses of up to \$71 billion for this reason. Regions with fewer resources to adapt are particularly affected. This can exacerbate economic inequalities between countries and within cities, demonstrating how climate change is not only an environmental challenge but also a challenge to global economic stability and justice.

The increase in hospitalizations due to heat-related illnesses significantly raises **healthcare costs**, as more resources are required to address emergencies and treatments related to heatstroke and other conditions. This places greater pressure on hospitals and emergency services and leads to increased operating expenses. Furthermore, this increased cost can affect the ability of public administrations to allocate resources to other areas of social welfare.

Tourism is one of Barcelona's key economic activities: it represents 14% of GDP and generates **more than 150,000 jobs**. Rising temperatures may discourage visits to the city as a tourist destination. Currently, and taking the entire Spanish Mediterranean coast as a reference, a decrease in the choice of this territory as a tourist destination has already begun to be observed, in favor of the Cantabrian coast of the Iberian Peninsula, according to the study "Socioeconomic and Labor Impact of Climate Change," by the Economic and Social Council of Spain (13).

Although a significant decrease in the number of annual visitors is not expected, a reduction in tourists could be observed during the warmer months, especially among those over 60, who are more vulnerable to the heat. This could lead to a less seasonal approach to tourism, with more visitors during other times of the year, thus easing the pressure on the city's infrastructure and services, such as water availability.

About the environment

The heat also affects the vegetation and wildlife in the city, disrupting **the dynamics of urban ecosystems**. With high temperatures, trees suffer water stress, a situation exacerbated by water scarcity, as the hottest periods often coincide with droughts.

Many trees die as a result of this combination of factors, and their replacement not only represents a significant public expense but also entails a loss of environmental quality. Young trees planted to replace those lost do not immediately offer the same socio-environmental benefits, such as shade, temperature regulation, or improved air quality.

The effects of heat on the environment also include a **reduction in biodiversity**. The progressive disappearance of native species and the **emergence of pests** and invasive species are being observed. In addition to urban wildlife, the well-being of companion animals must be considered, as their mortality increases during periods of extreme heat, especially among sick or elderly animals.

On the other hand, rising temperatures favor **changes in the distribution and proliferation of vectors and reservoirs of infectious agents**, such as mosquitoes and ticks, which can facilitate the transmission of diseases like dengue, Zika, and West Nile fever. The risk of gastrointestinal infections also increases due to water and food contamination caused by the direct effects of the heat.

Extreme heat **increases the risk of urban and forest fires**, as high temperatures and drought dry out vegetation and facilitate the spread of fire. This affects both urban areas and forested zones, endangering safety and infrastructure. Fires increase the economic burden and the need for emergency services to respond.

Finally, heat has a **direct impact on air quality**. High temperatures favor the formation of pollutants such as tropospheric ozone and cause an overall increase in air pollution. Likewise, pollen concentrations intensify and allergy seasons lengthen, worsening the symptoms and frequency of respiratory illnesses among the population.

The heat does not affect everyone equally.

The effects of temperatures can vary depending on several factors: individual (gender, age, economic status), temporal, and territorial. It is important to note, however, that these factors often cannot be reduced to a single characteristic, as people have multiple identities that interact and overlap. Thus, in the context of heat vulnerability, **intersectionality** implies that people are not equally vulnerable. People belonging to marginalized or disadvantaged groups may experience greater vulnerability to heat waves due to the interaction of various identities and social factors.

To effectively manage heat vulnerability, it is crucial to consider intersectionality and recognize that climate change mitigation and adaptation policies and measures must address existing inequalities. This means ensuring that solutions are inclusive and accessible to everyone, regardless of their identity or social status. Furthermore, addressing the underlying causes of inequality, such as homelessness, poverty, and discrimination, is essential to reducing heat vulnerability.

Individual factors

The impact of heat on mortality is not uniform: there are significant differences depending on various social and personal factors, such as gender, age, and socioeconomic status (see the 2030 Climate Emergency Action Plan). These axes of inequality determine vulnerability to extreme temperatures.

- **Gender:** In Barcelona, heat-related mortality risks are higher for women than for men. For example, in 2023, comparing the risk of death when the temperature reaches the intense heat threshold (34 °C) with respect to a comfortable temperature (24.9 °C), the risk for women increased by 75%, while for men it increased by 27%.

These differences may be due, on the one hand, to the social conditions of older women, who often live alone, and on the other hand, to differences in thermoregulatory and physiological mechanisms.

• **Age:** Older people are the most susceptible to the effects of temperature, and babies may be too. For example, in 2023, comparing the risk of death when the temperature reaches the intense heat threshold (34 °C) to that at a comfortable temperature (24.9 °C), the risk for people aged 75 and over increased by 56%, while for younger people it increased by 37%.

• **Socioeconomic status:** Economic inequality also plays a determining role. People in vulnerable situations have more difficulty protecting themselves from the heat, for example, due to a lack of air conditioning.

Figure 9.
Groups in situation
of vulnerability to
heat.

Source: Heat health
preparedness guidance –
health and community services.
State of Victoria, Australia,
Department of Health.

December 2022.

Grupos vulnerables al calor



Temporary factors

Vulnerability to heat can also vary over time. In recent decades, some studies have detected a slight decrease in the risk of heat-related mortality in summer. This improvement could be related to greater physiological acclimatization, changes in habits (such as wearing lighter clothing or staying better hydrated), and structural improvements such as the increase in air-conditioned homes, better housing quality, or the overall health of the population.

Territorial factors

Not all areas of the city experience the heat with the same intensity. Physical, urban, and social conditions vary within the same municipality, resulting in differentiated territorial vulnerability. The impacts of heat depend on multiple variables, such as the state of repair of buildings (which directly affects indoor thermal comfort), the presence and type of vegetation, the percentage of impermeable surfaces, urban density, the dispersion of heat sources, and the socioeconomic level of the neighborhood.

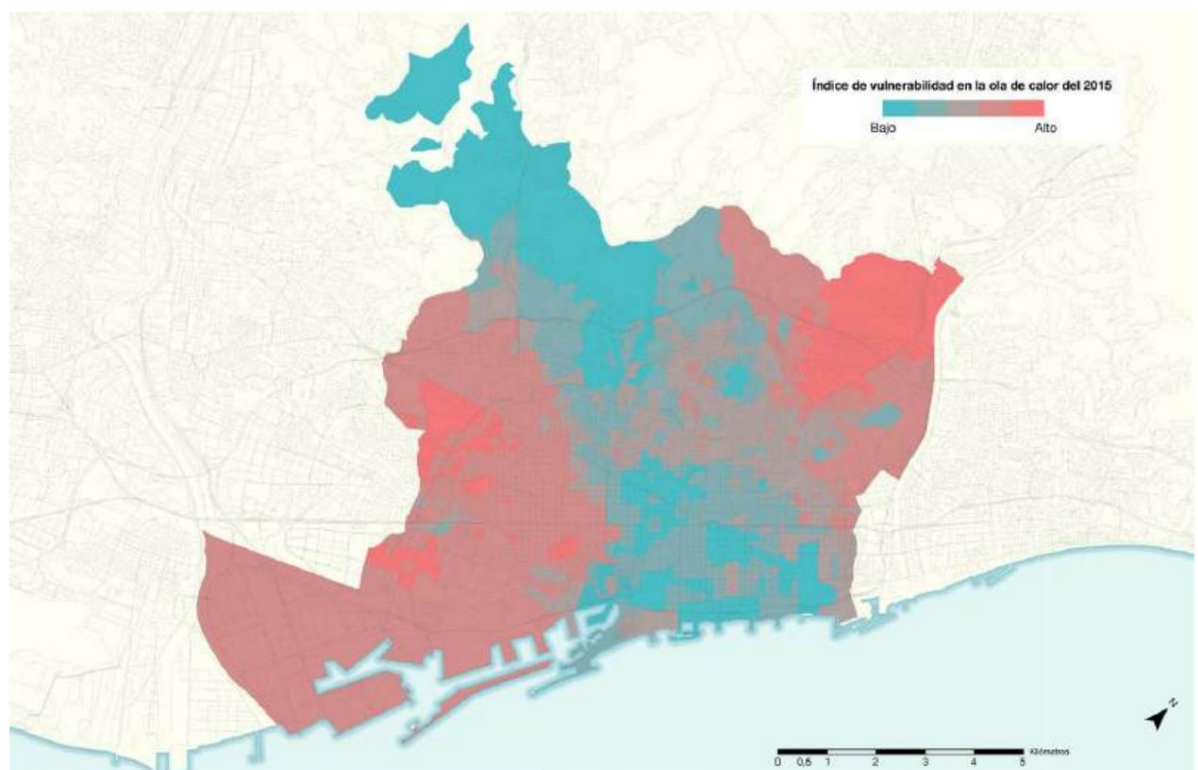
These differences make certain areas of the city particularly sensitive to episodes of extreme heat and require priority actions to protect the health and well-being of the people who live there.

In Barcelona there are several initiatives that analyze vulnerability to heat based on different factors.

2015 Heat Risk Map.

To develop the Climate Emergency Action Plan, the heat waves of 2012 and 2015 and their geographical distribution were analyzed. Simultaneously, a heat vulnerability map was created based on the analysis of four indicators: population over 75 years of age, building energy performance in relation to cooling demand, lack of vegetation, and the socioeconomic indicator of insufficient education. Finally, by cross-referencing the two maps, a heat risk map was obtained, showing that **the most affected areas of the city would be the Nou Barris district and some areas of the Sants-Montjuïc, Les Corts, and L'Eixample districts.**

Figure 10. Risk map in the 2015 heat wave, cross-referencing information on the heat experienced in 2015 in the territory with the different factors.



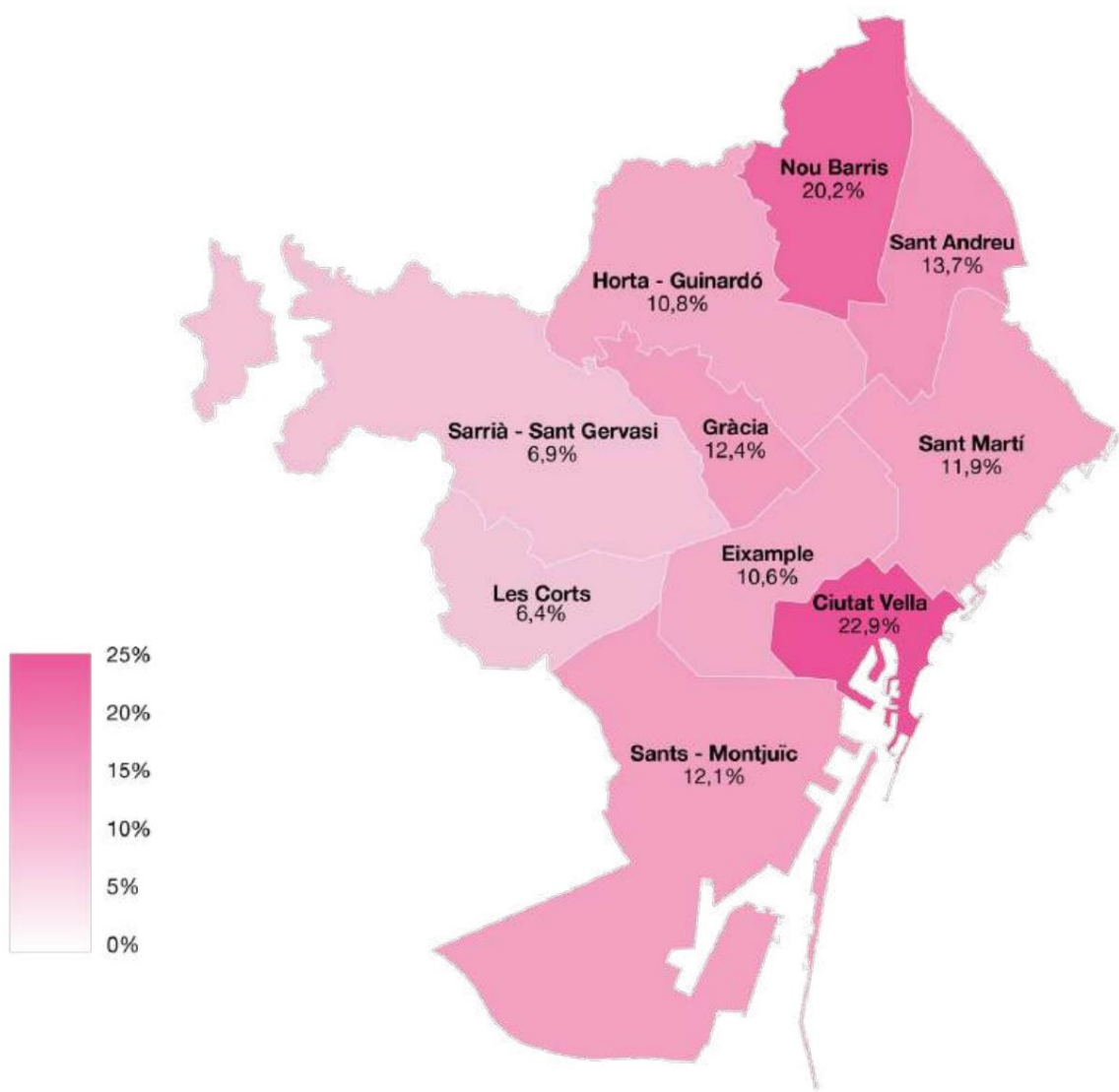
Source: Barcelona Regional 2017.

Report on residential insecurity in Barcelona 2017-2020

Another source of information is the survey conducted by the ASPB on energy poverty. The results indicate an increase in the number of residents struggling to keep their homes at a comfortable temperature in summer, particularly in the districts of Sants-Montjuïc, Ciutat Vella, Nou Barris, and L'Eixample.

Figure 11.
Percentage of
people who cannot
afford to maintain their
housing
with suitable
temperature during the
warm months by district.

Source: ASPB with data
from the Barcelona Health
Survey (ESB) 2021.



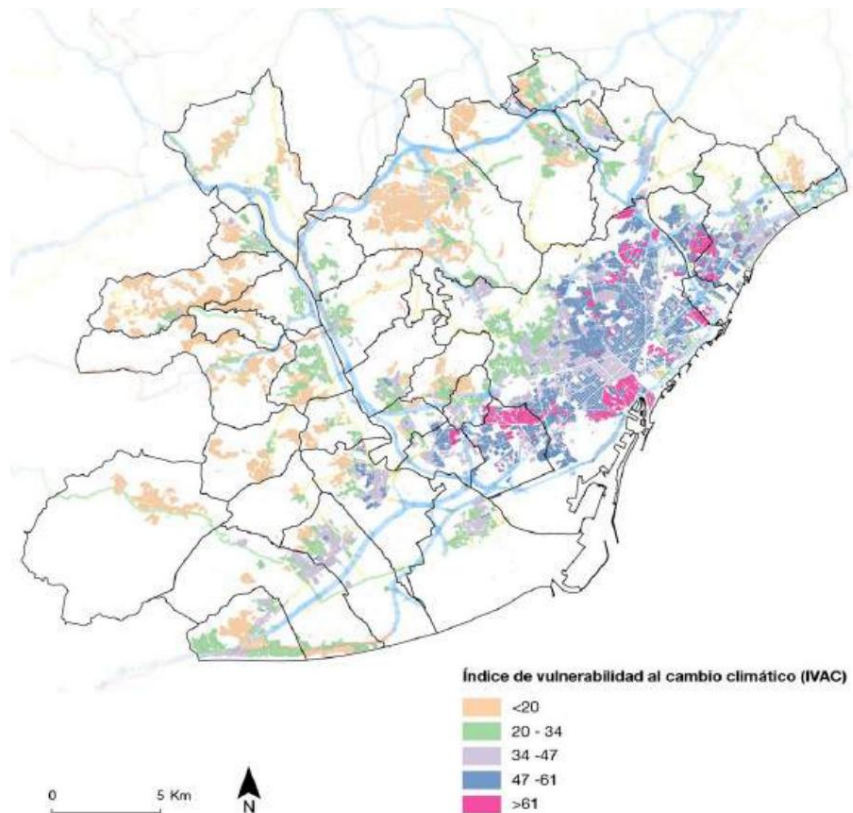
Climate vulnerability index in the metropolitan area

A study on vulnerability to climate change in the Barcelona metropolitan area has also identified vulnerability hotspots in relation to a set of climate indicators projecting temperature increases. This Climate Change Vulnerability Index (CVI) identifies areas and social groups most exposed to rising temperatures in the Barcelona metropolitan area by combining climate and socioeconomic data. According to this index, some 525,827 people, representing 16.13% of the metropolitan population, exhibit high vulnerability to climate change.

In the city of Barcelona, the highest vulnerability values to rising temperatures, with an IVAC >61, are located primarily in the neighborhoods of the districts of Ciutat Vella (especially Raval and Barceloneta), Nou Barris (large areas of Ciutat Meridiana, Prosperitat, and Turó de la Peira, but also other neighborhoods), Sants-Montjuïc (Sants-Badal, Sants, and Poble-sec), Sant Martí (Besòs i el Maresme), Sant Andreu (Trinitat Vella), Horta-Guinardó (Carmel and some areas of Can Baró), and Gràcia (Vila de Gràcia). These areas are therefore the priority for implementing adaptive measures.

Figure 12. Climate Change Vulnerability Index (CVI).

Source: Garcia-Sierra, M. [Marta]; technical support: ED [Elena Domene]; Andreu, M. [Marta]. (2022). Heat in the future: Climate Change Vulnerability Index (CVI)



New vulnerability index being developed within the framework of the project Climate Ready

Work is also underway to define a new climate vulnerability index through the **Climate Ready project**. In this regard, an innovative tool is being developed that integrates data such as climate shelters, shaded areas, energy consumption, and income, with the aim of improving support for citizens experiencing energy poverty and climate risk. This new vulnerability index is based on the integration and analysis of these indicators, allowing for the assessment of energy and climate vulnerability in Barcelona and its comparison across different areas of the city. The variables considered include energy consumption, climate data, socioeconomic data, and building characteristics.

**Strategy for
coping with
the heat.
Heat Plan
2025-2035**

**Where
are we?
Current
situation**



Strategic framework: alignment with the city's strategic plans

Barcelona is a city that, for the past 25 years, has been making increasingly ambitious commitments to address climate change and its effects. It has developed and implemented climate policies that, in many cases, have been pioneering and have served as examples for other major cities around the world. Recently, on November 29, 2024, the Governing Commission approved the **Climate Plan, a government measure** that compiles climate and sustainability measures aimed at mitigating the factors that contribute to climate change and reducing its effects, as well as increasing the city's resilience. This current Heat Plan is one of the six lines of action that make up the new Climate Plan.

The following are the plans and strategies most directly related to the Heat Plan:

The **2030 Climate Emergency Action Plan** : on which the MdG Climate Plan is based.

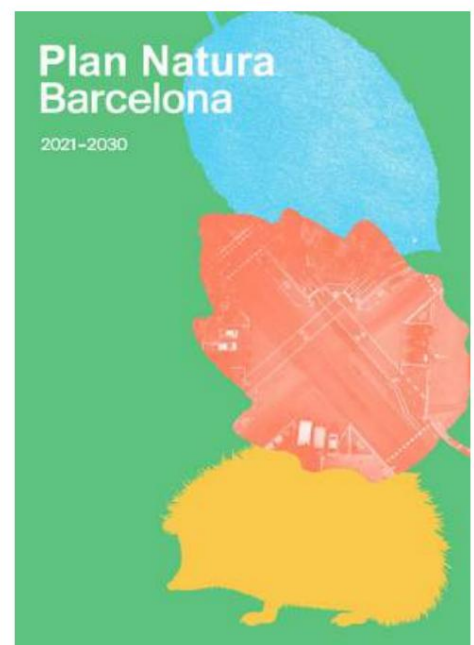
To design this Plan, climate projections and a vulnerability analysis were carried out, which, among other things, studied the urban heat island effect and how heat impacts other areas of the city. It has four lines of action: mitigation, adaptation and resilience, climate justice, and promoting citizen action.

It has a specific line of action on what it means to "guarantee thermal comfort" and many other related ones.



The **Natura Plan**: It emphasizes the importance of establishing a functional urban green infrastructure that maximizes social and environmental services, especially those related to health and adaptation to climate change, with two strategic lines of action closely linked to the Winter Heating Plan:

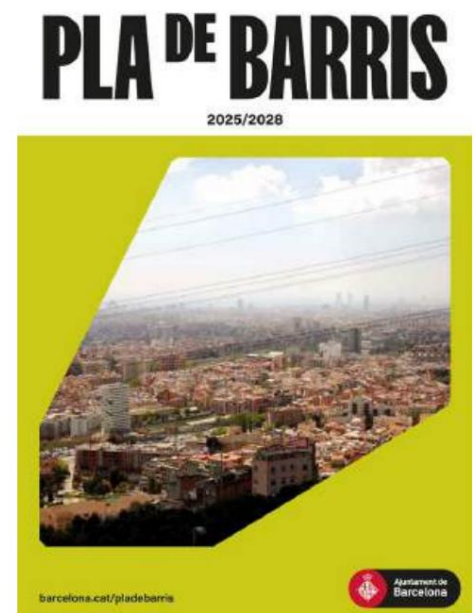
- More green spaces, an objective that was initially promoted through the Barcelona Climate Commitment, which pledges to increase Barcelona's green spaces by 1 m² /inhabitant by 2030.
- More naturalization of green spaces to maximize the contribution of ecosystem services in terms of thermoregulation, acoustic comfort, increased biodiversity, energy savings, reduction of CO₂ emissions and air pollution, regulation of the water cycle and production of food for people and wildlife.



The **Technical Plan for the Use of Alternative Water Resources in Barcelona (PLARHAB 2020)**: It includes an analysis of the alternative water resources available in the city and the different uses they can serve. In relation to the Heat Management Plan, all of this is essential to allow for the growth of green spaces in the city and other cooling areas. The PLARHAB also proposes tools to conserve and improve the ecological status of the water bodies surrounding the city.



The **Neighborhood Plan**: The third edition (2025-2028) of the Neighborhood Plan has as one of its objectives "adapting the city's public spaces and facilities to the new climate and social reality by enhancing green infrastructure and shaded areas and improving connectivity to bring them closer to the neighborhood." It incorporates several actions related to thermal comfort in vulnerable neighborhoods, such as increasing green space, resurfacing, creating shade, and providing climate shelters. It also focuses on certain groups that are particularly vulnerable, such as the elderly and migrants.



Citizens' Commitment for a More Sustainable Barcelona: This document embodies the will and aspirations of the Barcelona + Sustainable Network to achieve a more prosperous, inclusive, equitable, resilient, and responsible city with regard to the impacts of our lifestyle at both the local and global levels. It defines various collaborative projects to be undertaken with the Network and, specifically, includes one dedicated to heat, "Let's protect ourselves from the cold and the heat: let's offer private and public spaces as climate refuges."



The **“Let’s Change for the Climate” 2030 plan**: It aims to promote changes in the dominant culture and help develop and extend a new culture of sustainability that intensifies climate action in the city of Barcelona. It promotes awareness-raising and shared responsibility initiatives with the city’s various stakeholders.



Furthermore, the BCN Heat Plan 2025-2035 is closely aligned with the city’s strategies for health, social inclusion, and human rights. It complements the Barcelona City Health Plan, which establishes equity as a guiding principle for reducing health inequalities; the Plan for Addressing Homelessness, which ensures the rights of homeless people, one of the groups most vulnerable to extreme temperatures; the Municipal Strategy against Loneliness, focused on older adults and community support networks; and other measures to promote social action, which define the city’s social priorities. This intersectoral alignment, which is also linked to the Barcelona Pact for the Safeguarding of Human Rights, currently under collective development, is key to guaranteeing a coordinated, efficient, and equitable response to the impacts of heat on the population.

We're not starting from scratch. What have we done so far?

Barcelona has been working for some time on various measures to cope with high temperatures, especially with programs for vulnerable groups, such as the elderly, babies, people with chronic illnesses, and people with disabilities who have limited mobility and self-care. This section describes the actions Barcelona has taken in the past and continues to take to become a more heat-resilient city.

Climate shelter network

Since 2020, Barcelona has offered a Climate Shelter Network, which currently includes nearly 400 public spaces and facilities such as libraries, community centers, parks, and swimming pools to protect citizens—especially vulnerable groups—from extreme temperatures, both in summer and winter. In 2024, new spaces were added, such as municipal markets, and this year, 2025, a new type of shelter has been introduced: micro-shelters. These are smaller, short-stay spaces that also offer thermal comfort and information on the location of the nearest conventional shelters.

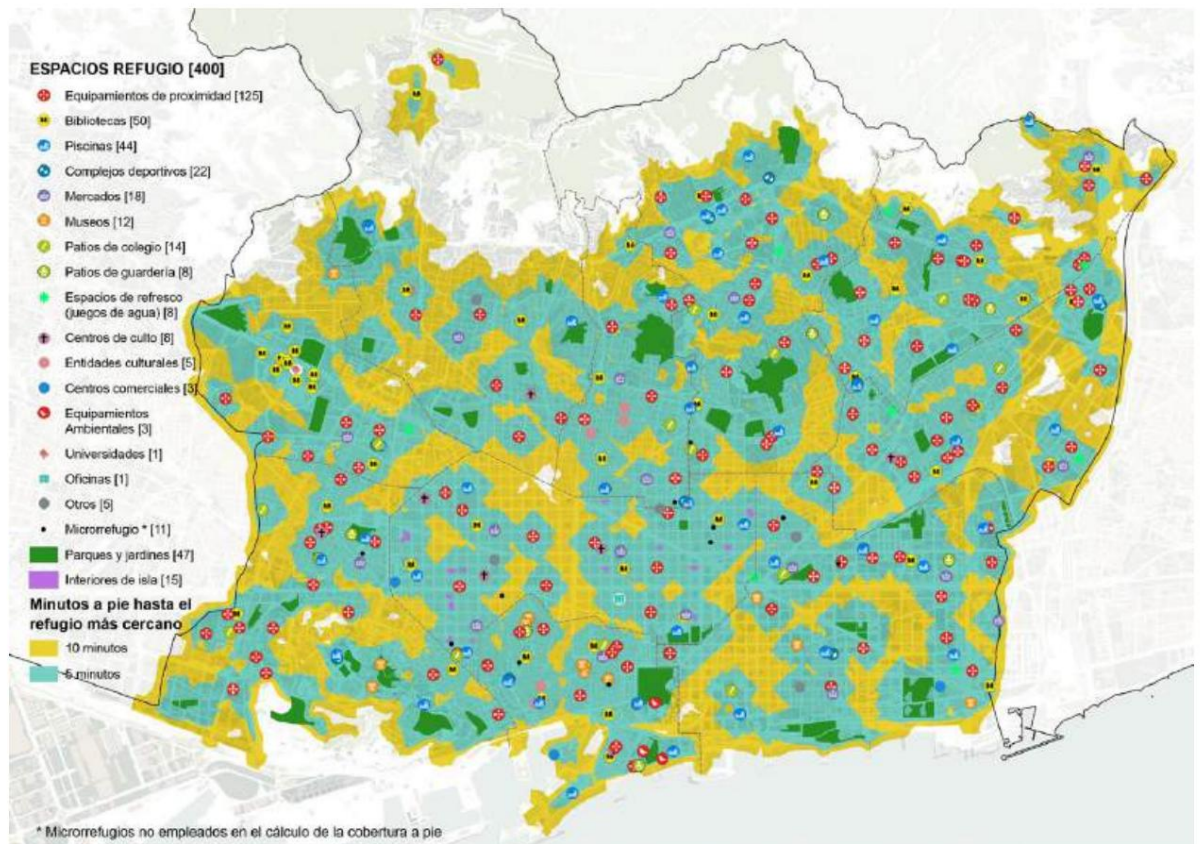
Climate shelters are free, accessible, safe spaces with rest areas and water, and they maintain a recommended temperature to ensure thermal comfort.

Currently, 99% of the population has a shelter within a 10-minute walk.

For more information: [Climate Shelter Network | Barcelona for the Climate](#) | [Barcelona City Council](#)

Figure 13.
Climate refuge space
network, June
2025.

Source: BR



Increased green space in the city

In recent years, Barcelona has deployed multiple initiatives to increase urban green space and strengthen its climate resilience: the Proximity and Interior Spaces Program (PEPI), which provides for the recovery of disused spaces; major city works (Glòries Park, Tretze Roses Park, new urbanizations such as those in the Marina del Prat Vermell neighborhood, major transformations such as La Sagrera, etc.); the redevelopment of major city thoroughfares (Via Laietana, La Rambla, Meridiana Avenue, etc.); new green roofs and party walls (public and private), new urban gardens and the promotion of private green spaces, among others.

The different planning instruments, such as the Natura Plan, the Master Plan for Trees or the Charter of Green and Biodiversity, foresee the increase of the green surface and define what this green should be like in order to maximize its ecosystem services: a functional and ecological green infrastructure, with equitable, accessible green, connected in the urban fabric and with the metropolitan green network.

In parallel, the Proximity and Interior Spaces Program (PEPI), with an investment of more than 150 million euros, will transform 357 urban spaces, adding 22.3 hectares of green space, 216 shaded areas and green roofs on municipal buildings.

These actions seek to strengthen ecological connectivity, improve environmental quality and adapt the city to the effects of heat.

Energy Advice Points

The Barcelona City Council offers Energy Advice Points (PAEs) in each district, where residents receive guidance on how to avoid service disconnections and ensure universal access to basic services. Information is also provided on reducing bills, improving energy efficiency, and accessing subsidies, including those related to solar energy. In summer, the PAEs intensify their outreach efforts to provide tips for beating the heat, both at home and outdoors, and continue to offer advice on energy saving and energy rights.

For more information: [Energy Advice Points | Energy Rights | Housing | Barcelona City Council](#)

Plan municipal de ola de heat

Since 2007, Barcelona has had a heat wave protocol, integrated into the Basic Civil Protection Plan, to protect public health during extreme heat. The Plan is activated through Civil Protection and coordinates municipal, social, and healthcare services to provide special assistance to vulnerable populations. The 2024 Municipal Heat Wave Plan includes measures such as activating climate shelters, identifying at-risk groups, specific actions like distributing fans, and strengthening personalized support. It is structured in three phases:

- **Preventive:** identification of people in vulnerable situations, dissemination of information, support for homeless people and action of social services and home care.
- **Alert:** reinforcement of support for people in vulnerable situations, activation of aid agreements and coordinated actions in the street.
- **Emergency:** intensification of measures, transfers to shelters and overnight stays at the CUESB in serious cases, with special attention to avoiding exposure during the hottest hours.

Training for professionals in the care sector

Every year, Barcelona City Council organizes training sessions on energy poverty and climate resilience for caregivers, such as those working in the Home Care Service. The aim is to equip them to identify situations of vulnerability and address the effects of heat, especially among people who do not have easy access to services like Home Care Support Programs or climate shelters. Since 2021, 700 caregivers have been trained through approximately 20 double sessions annually.

For more information: [Climate Justice | Barcelona for the Climate | Barcelona City Council](#)

Strengthening services for the elderly

During the warmer months, services for the elderly are reinforced with various initiatives, such as a telecare call campaign to provide basic guidance and facilitate access to services. Information campaigns are also conducted in assisted living facilities and sheltered apartments, and communal areas are opened as climate-controlled refuges. In senior residences, hydration is increased, and comfortable, temperature-controlled areas are prepared. Finally, the Home Care Service carries out an awareness campaign and distributes informational brochures to its users.

For more information:

[Telecare | Seniors | Barcelona City Council](#)

[Serviced housing | Senior citizens | Barcelona City Council](#)

[Residential care homes | Senior citizens | Barcelona City Council](#)

[Home care | Elderly people | Barcelona City Council](#)

“Let's transform the playgrounds” program

The Barcelona school playground transformation program, spearheaded by the City Council and the Education Consortium, aims to make these spaces more equitable, green, and accessible for the community. In addition to diversifying play activities and promoting co-educational relationships, many playgrounds are being integrated into the network of climate shelters. The interventions include adding play areas, planting trees, creating shade, and increasing the amount of natural ground. The plan aims to ensure that by 2030, all playgrounds in public schools will be healthy, co-educational, and open to the community. So far, 70 schools have participated in the program, with improvements such as the restoration of 1,000 m² of green space, the creation of 2,213 m² of shade, the planting of 74 trees, and the installation of 26 water points.

For more information: [Barcelona City Council](#)

School Climate Plan

In response to heat waves and the effects of climate change on schools, Barcelona is launching the School Climate Plan, with an investment of one hundred million euros between 2024 and 2029 to provide climate control in 170 educational centers. The project, managed by the Education Consortium and partly funded by the tourist tax, includes the installation of solar panels, efficient climate control systems, and the renovation of playgrounds according to sustainable criteria. Fifteen centers have already been renovated, including seven new schools. The plan is being implemented in two phases and will benefit more than 55,000 students and teachers.

For more information: [Barcelona School Climate Plan | Education | City Council Barcelona](#)

Cradles for the Climate

The **Bressols pel Clima (Nurseries for Climate)** strategy promotes 47 initiatives in Barcelona's nurseries to address the climate emergency through six key areas: greening, energy transition, thermal comfort, waste management, sustainable food, and environmental education. Its aim is to guarantee healthy environments for young children, promote structural changes in all nurseries, and reduce their ecological footprint. It includes self-diagnostic tools, training, and actions such as the naturalization of 25 courtyards and the creation of clothing exchange markets.

For more information: [Barcelona presents Bressols pel Clima, the municipal nurseries' strategy for climate justice | Nurseries](#)

Shelter schools

In 2018, the City Council received funding from Urban Innovation Action (UIA), a European Commission program, for the project “Adapting Schools to Climate Change through Green, Blue, and Grey,” which aimed to transform eleven schools into climate shelters. Interventions were carried out in eleven schools using a package of blue measures (adding water points), green measures (shaded spaces and vegetation), and grey measures (building improvements to enhance insulation). Children participated in designing climate solutions and evaluating the actions. The primary beneficiaries are the students and the wider community, as the schoolyards are open during the summer, allowing them to become part of the Climate Shelter Network.

For more information : <https://www.barcelona.cat/barcelona-pel-clima/es/escuelas-refugios-climaticos>

Shade in the city

Barcelona City Council is promoting a strategy to implement shade in public spaces with three priority lines: green infrastructure (trees and vegetation), permanent and productive structures such as photovoltaic pergolas, and temporary solutions such as shade sails for periods of intense heat.

By the end of 2024, 28 energy-generating pergolas that also provide shade have been installed, and a pilot program for shared self-consumption has been launched. The city currently has 108 permanent pergolas and 18 unique structures that provide shade and aesthetic value, in addition to 10 seasonal shade structures in parks such as Can Batlló and Can Robacols. There are currently 183 shaded children's play areas, 10 of which have awnings.

Last summer, Barcelona City Council also funded the installation of shade structures in 21 school playgrounds, with a budget of €300,000, creating a shaded area of 2,520 m². The schools were: Pere Vila in Ciutat Vella; Carlit in L'Eixample; Mossèn Jacint Verdaguer, Ramon Casas and Can Clos in Sants-Montjuïc; Duran i Bas in Les Corts; Tàber in Sarrià-Sant Gervasi; Josep Maria Jujol, Montseny and Pau Casals in Gràcia; Baloo, Mirades and Pau Casals in Horta-Guinardó; Antaviana, Calderón de la Barca, Mercè Rodoreda, Palma de Mallorca and Trinitat Nova (Sant Jordi school) in Nou Barris; Doctor Ferran i Clua and Mestre Enric Gibert i Camins in Sant Andreu; and Vila Olímpica in Sant Martí.

Recently, three ephemeral shade prototypes have been implemented in the Plaza Comercial, the Parc del Fòrum and the Rambla de Badal, with an investment of 300,000 euros, financed 80% by the City Council, to identify solutions that improve thermal comfort and health in urban areas where trees cannot be planted.

For more information: [The prototypes of the challenge launched by the City Council are already casting a shadow and Press release of the Shadows 2025 Program](#)

Permeable and reflective pavements

The Barcelona City Council is implementing various initiatives to reduce the urban heat island effect, most notably the use of reflective and permeable paving. Pilot tests have been carried out with **reflective paving** on the roofs of sheltered schools to improve albedo and reduce interior temperatures. The "**Panot of the 21st Century**" **competition has also been launched**, featuring a design that incorporates reflective properties, and **Heatland paving**, which has shown promising results in reducing surface temperature and noise, is being tested. Furthermore, the use of **permeable paving** is being explored as a measure to mitigate urban heat.

Sustainable urban drainage systems (SUDS) can also be considered a complementary solution, as they mimic natural drainage processes and have been successfully implemented in Barcelona since 2005. These systems help manage stormwater more sustainably. The city has awarded contracts to improve the maintenance of the SUDS, with preventive and corrective maintenance carried out throughout 2023.

For more information: [SUDS Technical Guide](#)

Green roofs, party walls and rooftop restoration

Roofs, walls, and party walls are considered productive surfaces with the potential to generate various socio-environmental services, such as renewable energy, urban green spaces, food crops, water harvesting, and social areas. Rooftops, which represent 67% of the city's roof area, are key to urban sustainability. The City Council promotes several initiatives to encourage green roofs and party walls, such as the Green Roof Competition, which has already awarded prizes to projects that have added urban greenery, renewable energy generation facilities, and spaces for wildlife. Furthermore, competitions and grants have been launched to promote the creation of new green roofs, contributing to the improvement of the urban landscape and biodiversity.

For more information: [Grants for green roofs and naturalization | Urban landscape](#)

Ornamental fountains and water features

Ornamental fountains use recirculating water and are part of urban planning solutions that bring beauty, richness and vitality to the environment, as well as being spaces to cool the atmosphere.

Maintenance of the 324 fountains already available in the city has been intensified, and environmental criteria have been incorporated into the fountain maintenance contract. In addition, the sources have been included in the groundwater supply network to reduce drinking water consumption.

Water features are cooling areas that use potable water without recirculation, but with low-consumption solutions. They do not use earthen springs or elements that can accumulate water or function as animal watering troughs.

They are equipped with accessible, non-slip, zero-depth paving and are removable during the off-season (from late June to early September). As of the closing date of this document, eight cooling areas are operational, four more than last year. Potential locations for new water features have been identified. Finally, a mobile application has been developed to help the public locate these facilities.

For more information: [Barcelona City Council](#)

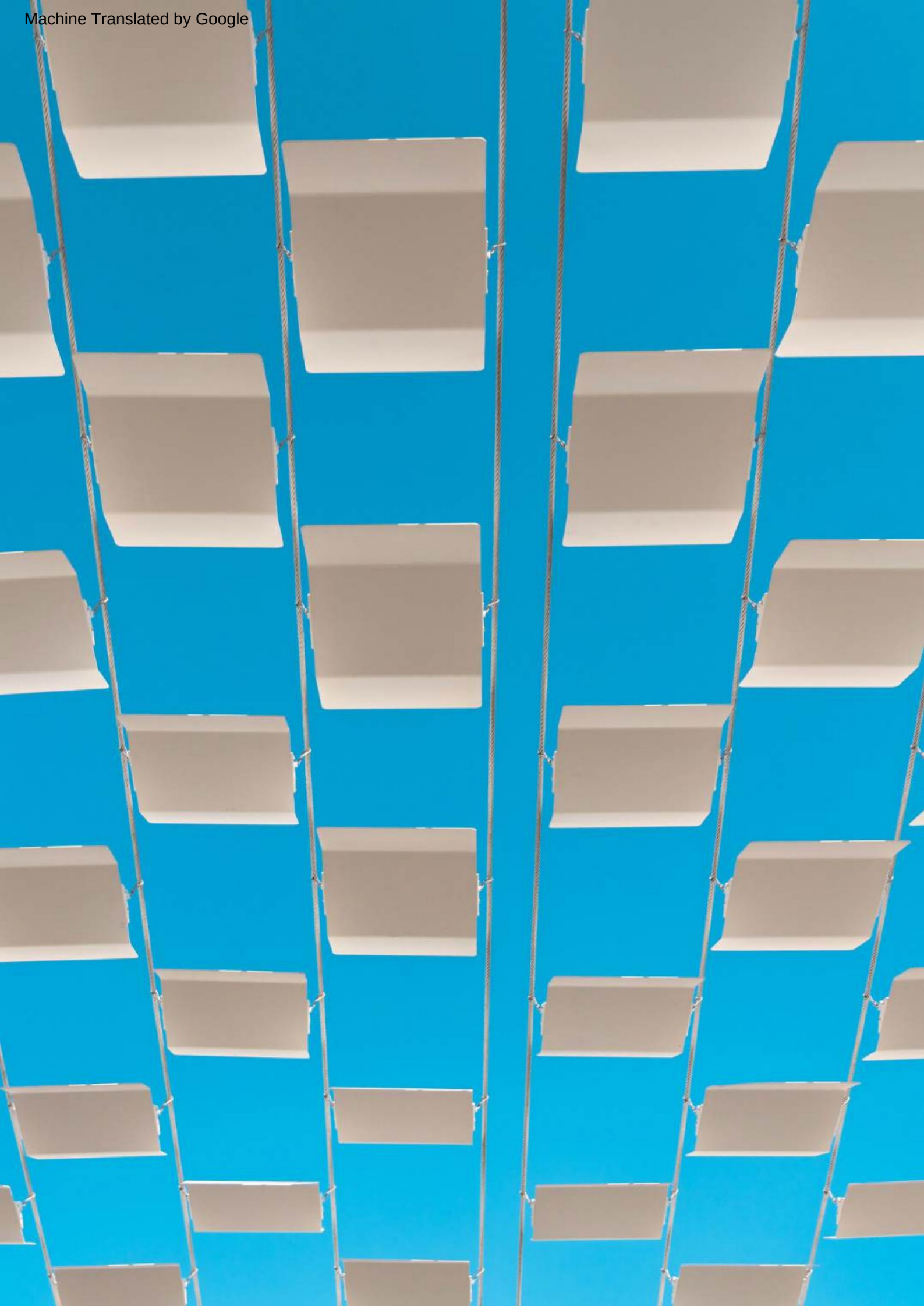
Drinking water sources

Barcelona has approximately 1,700 drinking fountains that provide water to pedestrians. All of these fountains share a common feature: a tap for drinking, regardless of their shape or material.

Since 2013, there has been a mobile app, Fonts BCN, that allows everyone to know at any time which fountain is closest, how to get there, and obtain basic information about all the drinking fountains. In addition to its public utility, the aim is to raise awareness and highlight this public drinking water supply network.

For more information: [Drinking and ornamental fountains | Urban planning, Transition Ecology, Urban Services and Housing](#)

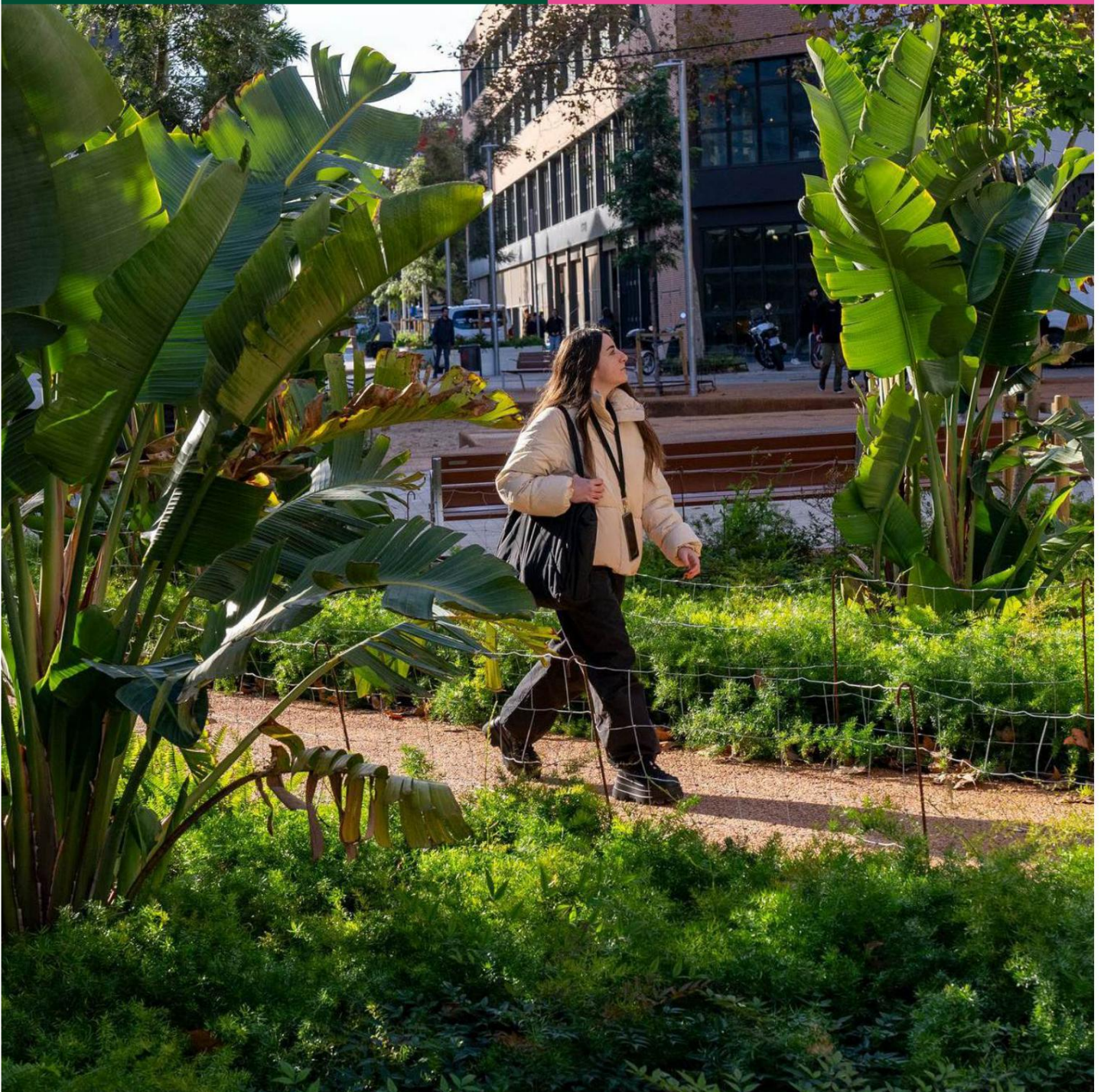




**Strategy for
coping with
the heat.**

**Heat Plan
2025-2035**

**Where
do we want to g**



Mission, vision and values

Vision

A prosperous and climate-resilient city that puts its citizens at the heart of its policies, with special attention to vulnerable neighborhoods and communities. A livable, fair, safe, and healthy city, with infrastructure capable of withstanding future extreme conditions and an empowered community able to respond collectively to the challenge of urban heat.

Figure 14.
The city we want.

Source: own elaboration.



Mission

To protect the health and well-being of all citizens from rising temperatures through advance planning, vulnerability reduction, integration of risk management into policies, and the promotion of a coordinated, evidence-based response. To ensure the city's continued functioning during extreme heat events and prepare it for the future.

Values

- **Climate justice:** which applies policies and measures taking into account the socioeconomic, gender, territorial and cultural diversity of its citizens and their intersectionalities; which facilitates their access, reduces inequalities and prioritizes interventions in those areas and groups in a situation of vulnerability to heat.
- **Sustainability:** to guarantee thermal comfort, it prioritizes nature-based solutions, passive solutions and, if necessary, uses the most efficient solutions with renewable energy.
- **Anticipation and prevention:** acting proactively, identifying and minimizing risks before they become emergencies. Advance planning is key to saving lives and ensuring essential services.
- **Risk management:** which incorporates the risk of heat into municipal preparedness, response and recovery protocols, ensuring a structured and coherent action in emergency situations.
- **Knowledge and scientific evidence:** which makes decisions based on scientific evidence and the precautionary principle, providing a detailed understanding of the impacts of heat, helping to prioritize resources and allowing interventions to be designed where they are most needed.
- **Innovation:** which drives applied research and the development of new, technological and social solutions to address the impacts of climate change and heat waves.
- **Collaboration and participation:** which promotes shared governance and involves all relevant actors (citizens, entities, public and private services), sharing responsibilities and actions.
- **Risk culture and awareness:** which educates and informs to raise awareness about the effects of heat and ways to protect oneself, both individually and collectively, encouraging the involvement of the whole of society.
- **Community resilience:** which fosters the collective capacity to prepare for, respond to and recover from heat waves and strengthens social bonds and co-responsibility among citizens, entities and institutions.

Goals

The Heat Strategy: The 2025-2035 Heat Plan aims to define a strategy for adapting Barcelona to the heat and protecting all citizens from its effects, paying particular attention to the protection of vulnerable groups. It is essential to integrate heat mitigation and adaptation into urban planning.

It subscribes to and adopts the objectives set out in the Climate Plan related to adaptation and climate justice that have to do with heat:

Table 4. Climate Plan objectives related to adaptation and climate justice regarding heat.

Source: Own elaboration

 Reduction of vulnerability to heat	 Warranty resources sufficient water	 Protection of the coast
• Increase the coverage of the climate shelter network so that 100% of the population has a shelter space within a 5-minute walk. • Improve the thermal comfort of 170 educational centers and 6 municipal residences for the elderly or people with disabilities. • Cover a minimum of 9 hectares with seasonal shade or structures fixed productive. • Provide all districts with a water play area.	• Achieve a maximum domestic consumption of drinking water of 90 liters/ inhabitant/day • Increase the use of groundwater by 2.7 hm3 . • Replace 100,000 m3 /year of drinking water for alternative water resources in municipal uses.	• Maintain a minimum beach width of 25 meters to ensure optimal functionality in terms of coastal protection and services.
 Reduced risk of flooding due to lack of drainage	 Increased services socio-environmental of the green	 Climate culture
• Implement 7 hectares of sustainable urban drainage systems (SUDS).	• Increase by 160 hectares (compared to 2015) and have 1 m2 more green space per inhabitant. • Create 10 biodiversity refuges. • Increase in 100 hectares the naturalized area. • Increase by 5% the tree cover (2037). • Increase from 30% to 40% of adapted tree species. • Unpave 3 hectares and recover organic soil.	• To ensure that 75% of citizens have adopted new values and habits to address the climate emergency.
		 Climate justice
		• To ensure that 0 people are in a situation of energy poverty.

Therefore, focusing on public space, buildings, infrastructure, and people, the following are considered our own objectives:

Figure 15. Objectives of the Heat Strategy.

Heat Plan 2025-2035.

Source: Own elaboration

Aumentar la resiliencia al calor del espacio público y las infraestructuras y reducir el efecto isla de calor

No podemos evitar el calor, pero sí podemos mejorar la resiliencia del espacio público y las infraestructuras que proveen de servicios básicos mediante la incorporación de soluciones de adaptación climática.



Mejorar el confort térmico de los edificios

Para garantizar el confort térmico de los edificios, sobre todo se necesitan acciones de rehabilitación energética. Para ello, hay que priorizar aquellas soluciones que no requieren consumo energético (soluciones pasivas) o, en el caso de que sea indispensable, que sean lo más eficientes posible y que utilicen energías renovables.



Proteger a los colectivos en situación de vulnerabilidad ante el calor

Los colectivos en situación de vulnerabilidad son los que más sufren los efectos del calor. Para protegerlos, hay que implementar acciones específicas en función de sus necesidades y particularidades, siendo especialmente sensibles a las interseccionalidades.



Mejorar las actuaciones y los protocolos de emergencia en caso de calor intenso

Durante las situaciones de emergencia derivadas de los episodios de calor en la ciudad, es indispensable que la respuesta de los servicios de emergencias y servicios sociales esté bien coordinada para poder garantizar su eficacia y eficiencia.



Avanzar hacia una nueva cultura del calor a través de acciones de comunicación, formación y participación

Sensibilizar a la población sobre los riesgos del calor y las medidas de adaptación mediante campañas informativas y educativas permite fomentar hábitos más responsables y una respuesta colectiva más efectiva ante los episodios de temperaturas extremas.



Impulsar la investigación, el conocimiento y la innovación sobre el calor

Con el fin de proteger a la ciudadanía y adaptar la ciudad al calor, hay que comprender mejor cómo el calor afecta a las personas, el espacio público y las infraestructuras.



Adaptar las políticas municipales y la gestión de los servicios municipales al calor

Hay que integrar el riesgo al calor y las medidas estructurales y operativas en los servicios municipales para mejorar la preparación, la respuesta y la recuperación durante los eventos de calor. También hay que adaptar la gobernanza para facilitar la toma de decisiones ante episodios de calor extremo.



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

We cannot prevent extreme heat or heat waves, but we can improve the resilience of public spaces and the infrastructure that provides basic services by incorporating climate adaptation solutions. Measures such as implementing shade structures, increasing green spaces, installing permeable and reflective pavements, and renaturalizing spaces help reduce temperatures in the city.

During periods of extreme heat, it is essential to guarantee the supply of water and electricity, mobility, and other vital services. It is necessary to analyze which infrastructures may be affected and plan how to prepare them, whether by replacing materials, improving interconnections, creating system redundancies, and so on. With this objective, actions are designed to protect these infrastructures.

To achieve this, the following lines of action will be implemented:

1. Incorporation of heat adaptation criteria into urban development projects
2. Definition of a plan for preventing damage to infrastructure due to situations of extreme heat
3. Reduction of direct heat emission sources
4. Increase in green space
5. Improved adaptation of greenery to heat
6. Implementation of the Plan for the recovery of green spaces affected by drought
7. Improvement of thermal comfort on the coast and of coexistence during overnight stays
8. Implementation of shadows
9. Construction of permeable pavements, SUDS and reflectors
10. Rationalization and optimization of ornamental fountains and improvement of public drinking water fountains
11. Installation of refreshment devices and new water features
12. Intensification of pest control in summer
13. Odor control and reduction

B. Improve the thermal comfort of buildings

Barcelona is an already built city. To guarantee the thermal comfort of buildings, energy-efficient renovations are essential. To achieve this, priority must be given to solutions that do not require energy consumption (passive solutions) or, if energy consumption is unavoidable, those that are as efficient as possible and utilize renewable energy. With this objective, measures such as the creation of green and reflective roofs on buildings and the revision of regulations concerning building construction and renovation are being designed.

To achieve this, the following lines of action will be implemented:

1. Promotion of the rehabilitation and renovation of homes to adapt them to the heat
2. Monitoring and development of new construction and rehabilitation regulations
3. Promotion of reflective and cooling covers
4. Development of a cooling plan for equipment used by citizens



Barcelona City Council

C. Protect vulnerable groups from the heat

Vulnerable groups suffer the most from the effects of heat. To protect them, specific actions must be implemented based on their needs and circumstances, paying particular attention to intersectionality. Thermal comfort measures are also planned for facilities that house some of these groups, such as schools and residences for the elderly or people with disabilities, among others, with interventions both inside buildings and in courtyards or outdoor spaces.

To achieve this, the following lines of action will be implemented:

1. Improvement and expansion of the climate shelter network
2. Improvement of thermal comfort in schools
3. Improvement of thermal comfort in facilities intended for the elderly and people with disabilities
4. Definition of a plan to improve the thermal comfort of the facilities
municipal services dedicated to serving vulnerable groups
5. Training for caregivers
6. Strengthening the PAE: energy rights against heat for all
7. Specific actions for the protection of people potentially in a situation of
vulnerability to heat
8. Expansion of services on beaches for people with reduced mobility
9. Protection of homeless people
10. Adaptation of events organized abroad
11. Extension of opening hours for parks and gardens

D. Improve emergency response procedures and protocols in case of intense heat

During heat waves in the city, it is essential that the response from emergency and social services be well-coordinated to ensure its effectiveness and efficiency. The review of this Action Plan and its potential improvements, as well as the optimization of communication channels with the public during heat waves, are some of the actions designed to achieve this.

To achieve this, the following lines of action will be implemented:

1. Improvements in internal protocols and contingencies for social care in situations emergency
2. Improvement of the alert system for weather events
3. Training of healthcare personnel
4. Conducting a simulation exercise in Barcelona at 50 °C

E. Moving towards a new culture of heat through communication, training and participation actions

Raising public awareness about the risks of heat and adaptation measures through information and educational campaigns fosters more responsible habits and a more effective collective response to extreme temperature events. Furthermore, shared responsibility and citizen participation are essential for the continuous improvement of this plan.

To achieve this, the following lines of action will be implemented:

1. Definition of an external communication plan
2. Promoting cultural change related to heat
3. Development of citizen participation and experimentation actions

F. To promote research, knowledge and innovation on heat

To protect citizens and adapt the city to the heat, we need to better understand how heat affects people, public spaces, and infrastructure. Detailed mapping of heat and its effects, along with corresponding vulnerability studies and analyses, is especially important. All this information will help prioritize resources and implement solutions where they are most needed. It is also necessary to test new solutions that efficiently improve adaptive capacity.

To achieve this, the following lines of action will be implemented:

1. Update of heat impact and vulnerability studies
2. Definition of a protocol for the system to monitor the effects of change climate in health
3. Monitoring of heat-attributable mortality
4. Improvement of thermal comfort monitoring systems
5. Establishment of knowledge synergies between research centers
6. Launching urban challenges to find innovative solutions
7. Feasibility study of different cold storage and production systems starting from renewable sources

G. Adapt municipal policies and the management of municipal services to the heat

Heat risk and structural and operational measures must be integrated into municipal services to improve preparedness, response, and recovery during heat events. Governance must also be adapted to facilitate decision-making in the face of extreme heat. Actions are planned to ensure that municipal services are better prepared to manage extreme heat events and to reduce risks and impacts.

To achieve this, the following lines of action will be implemented:

1. Improvement of thermal comfort in municipal buildings
2. Definition of a protection plan for municipal staff
3. Definition and implementation of internal communication actions
4. Activation of economic and financing tools for heat adaptation

Scope of the plan

Temporal scope

The effects of climate change are already tangible, and adapting to it requires profound changes in how we operate. Therefore, this Plan sets objectives for 2035 and defines actions for the short (2027), medium (2030), and long term (2035). It is worth noting that the climate projections on which this Plan is based are long-term (2050 and 2100).

The following criteria were taken into account when prioritizing actions:

1. Those aimed at improving people's health, especially those that are
They are in a vulnerable situation
2. The joint benefits that they entail
3. Ease of implementation (for example, if they are already in place or
(They involve the improvement of a pre-existing project, or if they are complex and require a cultural change)
4. The availability of resources to carry them out (both financial and human resources)

Other important criteria when defining and prioritizing actions are those described in the 2030 Climate Emergency Action Plan: they must provide a social return; be cost-effective, reproducible, flexible, scalable, co-produced, green or soft, and allow for action without compromising the future.

Territorial scope

Climate change knows no borders, and some issues have been analyzed considering geographical scales larger than the municipal level. On the other hand, in some cases it is necessary to go down to the district or neighborhood level, or, in some interventions, even to consider the urban microclimate.

Scope of competence

The Plan focuses on measures that can be implemented by the Barcelona City Council, but the Council cannot address all heat-related issues on its own. Therefore, beyond its own responsibilities, coordination with other supralocal administrations, such as the Government of Catalonia or the Metropolitan Area, is essential. Furthermore, some actions require collaboration with the private sector or shared responsibility from civil society. The Plan also anticipates the involvement of different sectors and areas of expertise and incorporates a holistic and cross-cutting approach.



Strategy for coping with the heat. Heat Plan 2025-2035

How will we do it? Lines of action



Lines of action

The following outlines the planned courses of action to achieve the established objectives. Each course includes the following information:

- **Objective** it addresses.
- **Name** of the line of action.
- **Status** of the action: it may be not initiated, initiated or consolidated.
- **Type of measure:** governance, normative or regulatory, operational and execution, economic, knowledge improvement and innovation, or communication and awareness.
- **Timeline:** short term (2027), medium term (2030) or long term (2035).
- **Duration:** continuous or punctual.
- **Description and actions:** explanation of what is planned in the line of action and its implementation. At the time of drafting this Plan, and based on the available knowledge, some lines of action are more defined and detailed than others.
- **Monitoring indicator:** some of the actions have their own monitoring indicators, beyond the percentage of execution of the action itself.
- **Milestone:** in some actions there are quantitative objectives to achieve.
- **Operational information:** regarding who the action is aimed at, which management is responsible for carrying it out, and other actors involved. Also, whether the action was proposed during the participatory process carried out.
- **Benefits:** Pending further information, the combined benefits of the actions are described qualitatively. These may include:

ÿ Environmental

- Reduction of temperature and the urban heat island effect: the implementation of green spaces, reflective surfaces and shade systems helps to lower the temperature in cities, making them more comfortable and resistant to heat waves.
- Reduction of energy or water consumption: improvements in building efficiency and sustainable urban planning decrease dependence on artificial cooling and optimize water use, contributing to the conservation of natural resources.
- Improved environmental quality: the decrease in air pollution, the reduction of greenhouse gas emissions and the increase in urban vegetation contribute to a healthier and more sustainable environment.

ÿ Social

- Co-responsibility: the involvement of the population and organizations in heat adaptation actions fosters a more resilient community, more aware and committed to the environment and sustainability.

- Improved well-being and health (mortality and morbidity): Reducing extreme heat through preventive measures decreases health risks, thus preventing illness and reducing mortality associated with heat waves.
- Reduction of inequalities: some measures such as guaranteeing access to cool spaces, providing aid, improving the comfort of public facilities, etc. make it easier for people in vulnerable situations to cope with the heat in better conditions.
- Increased resilience to heat: preparing infrastructure and communities to better cope with extreme heat allows for a more effective response to the effects of climate change.

• Economic

- Economic savings: energy efficiency and reduced resource consumption mean long-term savings for households, businesses and administrations.
- Job creation: the transition to a more sustainable infrastructure boosts sectors such as building renovation, green space management and renewable energy, generating new job opportunities.



Barcelona City Council



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Incorporation of heat adaptation criteria into urban development projects								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)			
				Duration	Continued Punctual			
Description and actions								
Incorporation of criteria for adapting to heat in public spaces (e.g., review of the instruction approved by Mayoral Decree for the environmental integration of projects and works, and updating of the Guide to Sustainable Urban Planning). Development of the protocol for transformative city projects (green form). These criteria may refer to the incorporation of vegetation and shade, reflective materials such as paving and highly reflective surfaces, sustainable urban drainage systems, water features such as fountains, ponds, or micro-sprinkler systems to cool the environment, and other urban design criteria that, for example, promote natural ventilation. Planning, development of information resources and delivery of sessions to train and empower those responsible for planning, drafting and reviewing urban development projects and providing them with resources. Establishment of the necessary mechanisms to ensure that these criteria are incorporated by the different promoters. Line of action linked to “Promotion of the rehabilitation and renovation of homes to adapt them to the heat”								
Monitoring indicators	Number of projects reviewed			Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility			Cost savings		✔
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation		
			Increased heat resilience		✔			
	Improvement of environmental quality		✔					
Operational information								
Target audience								
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)						
Other agents involved		Chief Architect's Office, Urban Services and Space Maintenance Management Public (Environmental Assessment and Management, Parks and Gardens), BIMSA, Neighborhood Plan, Districts						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Course of action		Definition of a plan to prevent damage to infrastructure caused by extreme heat.							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
					Duration	Continued Punctual			
Description and actions									
Development of a plan to adapt critical infrastructures to heat, which in its definition will include actions in the following areas and their interdependencies and possible cascading effects: - Electrical and energy network - Public transport - Mobility and road infrastructure - Water supply and sanitation - Telecommunications - Waste management The plan will be based on a vulnerability analysis to identify the infrastructure most vulnerable to the effects of extreme heat and its current characteristics. Using this information, along with climate data and projections (to understand extreme heat patterns and anticipate potential future changes), the technical measures to be implemented, the execution schedule, and the necessary resources will be defined. To this end, relevant working groups will be established, along with avenues for collaboration with the various public and private administrations and entities that provide the different services.									
Monitoring indicators				Milestone					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
	Lower energy or water consumption			Improved well-being and health (mortality and morbidity)			Job creation		
	Improvement of environmental quality			Reduction of inequalities					
				Increased heat resilience					
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Area Management of Mobility, Infrastructure and Urban Services (Office of the Chief Engineer)							
Other agents involved		Area Management of Mobility; Urban Infrastructure and Services (Barcelona Energy Agency, Barcelona Water Cycle, Urban Space, Cleaning and Waste Management); Management of the Municipal Institute of Informatics; Barcelona Municipal Services; Management of Mobility, Infrastructure and Urban Services; Metropolitan Transport of Barcelona (TMB); service providers; Barcelona Metropolitan Area (AMB); ADIF; GENCAT; ATM (Metropolitan Transport Authority of the Barcelona Area)							



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Reduction of direct heat emission sources								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
				Duration	Continued Punctual			
Description and actions								
Identification of existing heat sources and planning of measures to reduce their negative impact on the thermal comfort of public spaces.								
Regarding the existence of air conditioners at low heights: review the regulations and assess the implementation of control measures and aid for their relocation.								
Regarding the traffic of internal combustion engine vehicles: develop actions to promote sustainable mobility within the framework of the Urban Mobility Plan 2024.								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings		
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption	✔	Reduction of inequalities			Job creation		✔
			Increased heat resilience		✔			
	Improvement of environmental quality	✔						
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yeah						
Responsible area		Area Management of Mobility, Infrastructure and Urban Services; Management of Urban Planning; Management of Urban Services and Maintenance of Public Space; Districts						
Other agents involved		Resource Management and Digital Transformation (Communication)						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Increase in green area							
State	Not started Started Consolidated	Type of measurement	Governance	Calendar	Short term (2027)		
			Regulatory or normative		Medium term (2030)		
			Economic	Duration	Long term (2035)		
			Improved knowledge and innovation		Continued		
			Communication and awareness		Punctual		
Description and actions							
Increase the green area according to the objectives of the Natura Plan and reach an increase of 160 hectares of green space by 2030.							
Adapting public spaces and the urban fabric to the challenges arising from the climate emergency and the scarcity of green spaces and permeable soil. To this end, the existing potential of the urban fabric will be analyzed, urban quality will be redefined, and public spaces will be improved.							
Within the framework of the Shade Plan and the Proximity and Interior Spaces Program, a specific subprogram will be developed to plant 250 new trees in those priority spaces according to the studies of sun exposure and the vulnerability of the environment.							
Monitoring indicators	Green surface			Milestone	To have 160 hectares of green space by 2030		
Benefits		Environmental			Social		Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings	✔
			Improved well-being and health (mortality and morbidity)		✔		
	Lower energy or water consumption		Reduction of inequalities		✔	Job creation	
	Improvement of environmental quality	✔	Increased heat resilience		✔		
	Operational information						
Target audience		Citizenship					
Action proposed in the participatory process		Yeah					
Responsible area		Urban Planning Management					
Other agents involved		Urban Services and Public Space Maintenance Management (Parks and Gardens), Chief Architect's Office, Neighborhood Plan					



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Improve the adaptation of greenery to heat								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness		Duration	Continued Punctual		
Description and actions								
Establish and implement a long-term experimental design for the trees in the city of Barcelona. Monitoring should allow us to understand the health status of the trees under changing environmental conditions and their potential to provide essential ecosystem services. This knowledge will inform and support scientifically based decisions regarding the management of urban trees.								
Monitoring indicators	Number of tree species adapted to heat			Milestone	Increase the percentage of adapted tree species from 30% to 40%.			
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility			Cost savings		✔
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation		
	Improvement of environmental quality	✔	Increased heat resilience		✔			
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yesh						
Responsible area		Urban Services and Public Space Maintenance Management (Parks and Gardens)						
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability), research centers						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Implementation of the Plan for the recovery of green spaces due to drought								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)			
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual			
Description and actions								
Restore vegetation lost due to drought by redesigning urban green spaces to make them more resilient. In the event of drought, allow for tree irrigation to prevent irreparable losses. Continue planting until reaching 9,000 trees and palms by the end of 2025 (when there is no drought, the replacement rate is 2,000 to 2,500 trees per year) and restore 10 hectares of vegetation. Expand the groundwater network to ensure the necessary irrigation of parks and gardens, in accordance with the Alternative Water Resources Plan, to guarantee their survival, as well as the ecosystem benefits they provide.								
Monitoring indicators	Number of trees (ut) and green area in rest (m2)			Milestone	Replace 9,000 trees (2024-2026)			
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption		Reduction of inequalities			Job creation		
	Improvement of environmental quality		Increased heat resilience					
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Parks and Gardens)						
Other agents involved		Urban Services and Public Space Maintenance Management (Barcelona Water Cycle)						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Course of action	Improved thermal comfort on the coast and improved coexistence during overnight stays on the beaches							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
					Duration	Continued Punctual		
Description and actions								
Improvement of thermal comfort in the transition space between the city and the beaches with the provision of shade (with more vegetation and pergolas where planting is not possible) and the installation of drinking fountains in specific locations, and improvement of accessibility to the beaches to cool off, etc. Conduct a diagnostic study on beach use at night (including data on the number and types of users and on coexistence) by the nighttime beach management working group. Develop proposals to improve coexistence during nights with higher temperatures in the city. Develop a pilot project in 2026 with the prioritized proposals and actions. Strengthen communication of existing services and their schedules (beach bars, lifeguards, etc.).								
Monitoring indicators					Milestone			
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption		Reduction of inequalities			Job creation		
	Improvement of environmental quality		Increased heat resilience					
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Water Cycle)						
Other agents involved		Management of Social Rights, Health, Cooperation and Community; Management of Security, Prevention and Coexistence; Chief Architect's Office; Urban Services Management and Public Space Maintenance (Climate Change and Sustainability Office)						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Implementation of shadows							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)		
				Duration	Continued Punctual		
Description and actions							
Definition and implementation of the Shade Plan 2030. Based on the diagnosis of the radiation level of the different public spaces of the city, the criteria, milestones and objectives are defined to guarantee shade in most of the public space during the months of highest solar radiation and thus improve climate comfort. A series of short- and medium-term actions are planned to advance this objective and implement and test different solutions, thereby refining the technical and implementation criteria. Actions related to the implementation of shade structures proposed through participatory budgeting will also be considered, and those approved will be developed. The primary shade is provided by vegetation. Where planting is not possible, the second priority is fixed, productive, energy-generating pergolas. Finally, seasonal shade structures are planned.							
Monitoring indicators	Number of shadows implemented		Milestone	The specific objectives for 2027 are to install shade systems in 217 spaces, covering 52,000 m². The shade structures will be installed in playgrounds, schoolyards, and urban areas. Photovoltaic pergolas are not included in this list.			
Benefits		Environmental		Social		Economic	
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility			Cost savings	
			Improved well-being and health (mortality and morbidity)		✔		
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation	✔
	Improvement of environmental quality	✔	Increased heat resilience		✔		
Operational information							
Target audience		Citizenship					
Action proposed in the participatory process		Yeah					
Responsible area		Chief Architect's Management					
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability, Urban Space, Barcelona Energy Agency, Parks and Gardens), Neighborhood Plan					



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Construction of permeable pavements, SUDS and reflectors									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)				
				Duration	Continued Punctual				
Description and actions									
Promoting the permeability of large green spaces in the city and open spaces without facilities, such as school playgrounds. Replacing asphalt with permeable paving in some areas. Following an initial pilot test using Heatland reflective paving, a second pilot test with similar characteristics will be conducted on Martí i Franquès Street. One section will be paved with reflective asphalt, and another with conventional asphalt for contrast. Subsequent monitoring will be carried out. Additionally, the reflective effect of materials other than Heatland is being studied, and the construction of reflective paving based on recycled glass is planned. Depending on the results, a study will be conducted to assess the feasibility of implementing similar measures throughout the rest of the city.									
Monitoring indicators	Permeable surface area (ha)		Milestone	Target for 2030: depaving of 3 hectares, recovery of organic soil and implementation of 7 hectares of sustainable urban drainage systems (SUDS).					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		✔	Shared responsibility			Cost savings		
	Lower energy or water consumption		✔	Improved well-being and health (mortality and morbidity)		✔	Job creation		
	Improvement of environmental quality		✔	Reduction of inequalities		✔			
			✔	Increased heat resilience		✔			✔
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Area Management of Urban Planning and Housing, Management of Urban Services and Maintenance of Public Space (urban space)							
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability, Parks and Gardens, Barcelona Water Cycle); Consortium of Barcelona Education; Area Management of Culture, Education, Sports and Life Cycles; Neighborhood Plan; Chief Architect's Office							



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Course of action		Rationalization and optimization of ornamental fountains and improvement of public drinking water fountains				
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)	
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual	
Description and actions						
Analysis of the current state of ornamental fountains: following the shutdown of fountains due to drought, it is proposed to study the city model in terms of ornamental fountains, taking into account the cost of restarting fountains if they have been empty for a long period of time (which may be increasingly frequent). Conducting a study on the type of sustainable fountain that the city wants to promote in a context of climate change (water features, large surfaces, naturalized fountains, etc.). Assessment of the implementation of improvements to ornamental fountains through the Drought Recovery Plan and the Comprehensive Maintenance Plan (PMI). Improvement of maintenance, repair and segregation of the groundwater network of existing drinking water sources that require it through the PMI. Identifying locations where new drinking fountains need to be installed in public spaces and in facilities serving vulnerable groups, such as schools and daycare centers. Installing water fountains and bottle refill stations in public facilities and establishing partnerships with private entities to promote the use of free water fountains in private facilities (shopping centers, sports centers, etc.).						
Monitoring indicators	Number of improved ornamental fountains Number of improved drinking water sources		Milestone			
Benefits		Environmental		Social		Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings
			Improved well-being and health (mortality and morbidity)			
	Lower energy or water consumption		Reduction of inequalities			Job creation
	Improvement of environmental quality		Increased heat resilience			
Operational information						
Target audience		Citizenship				
Action proposed in the participatory process		Yeah				
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Water Cycle)				
Other agents involved		Chief Architect's Office; Area Management of Culture, Education, Sports and Cycles Life; Districts; Resource Management and Digital Transformation; Education Consortium Barcelona; private entities				



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Installation of cooling devices and new water features								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
					Duration	Continued Punctual		
Description and actions								
A pilot test will be conducted to install a cooling device consisting of a shaded seating area (pergola) with water dispensers. A subsequent study will analyze the results regarding public use, any instances of vandalism, maintenance needs, temperature reduction, etc. This information will inform the placement of these types of devices according to the priority areas identified in the Shade Plan.								
A pilot seawater refreshment area will also be installed on the waterfront to test its operation, its use and the necessary maintenance, taking into account the specific saline conditions of seawater.								
Definition of a proposed location for new features and creation of a city map of water features (which also serve as climate refuges), taking into account all planned actions on the territory from different areas and scopes of action (Neighborhood Plan, Heat Plan, district actions, etc.). Analysis and rationalization of new proposals based on needs, considering sustainable growth in relation to consumption (it must be kept in mind that wastewater is used).								
Assessment of the maintenance plan and maintenance costs to include it in future maintenance contracts for sources (assessment of the creation of a team specialized in the maintenance of said devices).								
Monitoring indicators	Number of new water features installed Number of refresh devices			Milestone	One water feature per district			
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		✔	Shared responsibility			Cost savings	
				Improved well-being and health (mortality and morbidity)		✔		
	Lower energy or water consumption			Reduction of inequalities		✔	Job creation	
				Increased heat resilience		✔		
Improvement of environmental quality		✔			✔			
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Cycle of Water), Neighborhood Plan						
Other agents involved		Districts; Management of Social Rights, Health, Cooperation and Community; Management of Urban Services and Public Space Maintenance (Climate Change Office and Sustainability, Urban Space); Chief Architect's Management						



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Intensification of pest control in summer									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness			Calendar	Short term (2027) Medium term (2030) Long term (2035)		
						Duration	Continued Punctual		
Description and actions									
Strengthening public works teams to intensify urban pest surveillance and control by providing more human and technical resources to municipal services responsible for pest control and implementing specific measures, such as structural modifications to storm drains to prevent mosquito breeding. Collaboration with citizens will also be promoted through information and awareness campaigns in public and private spaces on best practices for preventing pest infestations.									
Monitoring indicators					Milestone				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption			Reduction of inequalities			Job creation		
	Improvement of environmental quality			Increased heat resilience					
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Area Management of Social Rights, Health, Cooperation and Community							
Other agents involved		Resource Management and Digital Transformation (Communication)							



A. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect

Line of action: Odor control and reduction									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness			Calendar	Short term (2027) Medium term (2030) Long term (2035)		
						Duration	Continued Punctual		
Description and actions									
Analysis of the IRIS map for odors emanating from sewers and monitoring of actions taken and their effectiveness within the framework of the Odor Table. Study of other potential market techniques to reduce odors. Implementation of pilot tests in the areas of sewers, waste, and other emerging sectors to evaluate their cost-benefit ratio, such as the installation of chimneys and filters.									
Monitoring indicators					Milestone				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
	Lower energy or water consumption			Improved well-being and health (mortality and morbidity)			Job creation		
	Improvement of environmental quality			Reduction of inequalities					
				Increased heat resilience					
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		No							
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Water Cycle)							
Other agents involved		Management of Social Rights, Health, Cooperation and Community; Management of Services Urban and Public Space Maintenance (Cleaning and Waste Management, Evaluation and Management and Environmental)							



B. Improve thermal comfort in buildings

Action line: Promotion of the rehabilitation and renovation of homes to adapt them to the heat								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Improved knowledge and innovation Communication and awareness		Duration	Continued Punctual		
Description and actions								
Develop a comprehensive strategy that combines financial aid (direct subsidies, tax breaks, etc.), regulation, advice, technical support and assistance to owners and neighborhood communities and citizen awareness to ensure that the city's homes are more heat-resistant, more thermally comfortable and more efficient. Empowering citizens and organizations to access rehabilitation aid. Develop a guide with thermal comfort criteria for the design and construction of buildings. Line of action linked to the “Incorporation of heat adaptation criteria in urban development projects”.								
Monitoring indicators	Number of aid applications			Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings		✔
			Improved well-being and health (mortality and morbidity)		✔			
	Lower consumption energetic or water	✔	Reduction of inequalities		✔	Job creation		✔
	Improvement of environmental quality	✔	Increased heat resilience		✔			
Operational information								
Target audience		Citizens and construction companies						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Planning and Housing Management						
Other agents involved		Urban Services and Public Space Maintenance Management (Energy Agency of Barcelona, Office of Climate Change and Sustainability); Neighborhood Plan						



B. Improve thermal comfort in buildings

Line of action: Monitoring and development of new construction and rehabilitation regulations						
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)	
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual	
Description and actions						
Monitoring the evolution of regulations relating to energy efficiency and renewable energy generation in buildings and, where appropriate, adapting municipal regulations and instructions to changes in European and state regulations with the aim of optimizing their applicability and alignment with municipal energy improvement milestones. Assessment of the possible development of new regulations or revision of existing ones at the city (ordinances) or municipal (instructions) level.						
Monitoring indicators			Milestone			
Benefits		Environmental		Social		Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings
			Improved well-being and health (mortality and morbidity)			
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation
			Increased heat resilience			
	Improvement of environmental quality	✔			✔	
Operational information						
Target audience		Citizenship				
Action proposed in the participatory process		Yeah				
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Energy Agency)				
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)				



B. Improve thermal comfort in buildings

Line of action: Implementation of cooling roofs and walls									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
					Duration	Continued Punctual			
Description and actions									
Promote productive roofs, facades, and walls (providing environmental or other benefits) for single or preferably mixed use, such as capturing renewable energy, green roofs, cistern roofs, or, failing that, cool (reflective) roofs. Specifically, the plan is to create ten green roofs and twenty green party walls. Consider incorporating productive cover requirements into planning regulations. To offer resources and guides so that owners can implement these solutions and thus contribute to a cooler and more sustainable city. Promotion and financing: development of tax incentives, subsidies or aid to facilitate implementation. Monitoring of environmental and social impact: evaluation of improvements in thermal comfort, reduction of energy consumption (air conditioning), contribution to sustainability and climate resilience of the city. Conduct a pilot test on reflective roofs. If the results are favorable, and after also evaluating the impact on the surroundings (environmental, social, and landscape), its implementation will subsequently be extended to more buildings, prioritizing according to location, vulnerability of the users, needs, and solar exposure (the order mentioned above will be taken into account).									
Monitoring indicators			Milestone						
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption			Reduction of inequalities			Job creation		
				Increased heat resilience					
	Improvement of environmental quality								
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Urban Planning and Housing Management, Urban Services and Maintenance Management Public Space (Climate Change and Sustainability Office)							
Other agents involved		Chief Architect's Office; Urban Services and Space Maintenance Management Public (Barcelona Energy Agency, Parks and Gardens); BIT HABITAT							



B. Improve thermal comfort in buildings

Line of action: Development of a cooling plan for equipment used by citizens

State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)
				Duration	Continued Punctual

Description and actions

Development of an investment program in cooling systems or other systems, such as additional ventilation or active and/or passive systems that improve thermal comfort in municipal facilities for citizen use that only have heat production and distribution systems (winter), but not cooling systems (summer).

Special attention will need to be paid to facilities that serve as climate shelters (this plan is in addition to the Schools Climate Plan and the Markets Climate Plan, which are already being implemented). Climate shelters will be prioritized to ensure they can fulfill this function in a severe future climate scenario. This will be achieved using updated temperature databases and short- and medium-term forecasting models.

Study the facilities that could be converted into climate shelters and assess the investment estimates needed to adapt them.

Monitoring indicators	Investment made (euros)	Milestone	
	Buildings under intervention		

Benefits		Environmental		Social		Economic
	Reduction of temperature and the urban heat island effect	✓	Shared responsibility		Cost savings	
	Lower energy or water consumption		Improved well-being and health (mortality and morbidity)	✓		
	Improvement of environmental quality		Reduction of inequalities	✓	Job creation	
			Increased heat resilience	✓		

Operational information

Target audience	Citizenship
Action proposed in the participatory process	No
Responsible area	Urban Services and Public Space Maintenance Management (Energy Agency of Barcelona), General Services Management
Other agents involved	Area Management of Culture, Education, Sports and Life Cycles; Barcelona Education Consortium



C. Protect vulnerable groups from the heat

Line of action: Improvement and expansion of the climate shelter network								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)			
	Duration		Continued Punctual					
Description and actions								
Consolidation and improvement of the network of climate refuge spaces to provide thermal comfort to the population. These can be indoors and outdoors, public and private, for long and short stays, and free of charge (except for municipal swimming pools, which offer reduced prices for vulnerable groups). The improvements should be incorporated in various areas: - Extended hours to ensure greater coverage, especially in August and in more vulnerable areas. - Incorporation of new spaces through collaborations with other administrations, associations and private and neighborhood entities. - Improvement of facilities and services (e.g., drinking fountains, comfortable seating, etc.). - Adaptation to some specific groups (e.g., spaces for babies and children, etc.). - Conducting internal quality audits to verify the suitability of the spaces. - Programming of environmental education and training activities on climate change, heat and resilience in some of these spaces. - Improvements in communication to publicize the network, to reach all groups, especially those in vulnerable situations, as well as when it is most advisable to use these spaces. - Implementation of campaigns to incorporate private spaces into the network, especially short-term micro-shelters. - Incorporation of the necessary requirements to be a climate refuge in the new tenders that are carried out. - Improvements in the safety of public shelter spaces to guarantee the well-being of vulnerable groups, such as the homeless.								
Monitoring indicators	Percentage of population coverage within 10 minutes and 5 minutes walk		Milestone	Having a climate shelter within a 10-minute walk				
Benefits		Environmental		Social		Economic		
	Reduction of temperature and the urban heat island effect		Shared responsibility		✓	Cost savings		
			Improved well-being and health (mortality and morbidity)		✓			
		Lower energy or water consumption		Reduction of inequalities		✓	Job creation	
		Improvement of environmental quality		Increased heat resilience		✓		
Operational information								
Target audience		Citizens and populations vulnerable to extreme temperatures						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)						
Other agents involved		Districts; Library Consortium; Area Management of Culture, Education, Sports and Life Cycles; Management of Urban Services and Maintenance of Public Space (Parks and Gardens, Barcelona Water Cycle); Barcelona Education Consortium; Generalitat of Catalonia; Metropolitan Area of Barcelona; Barcelona Provincial Council; shops, museums, places of worship, City Development (Neighborhood Plan), etc.						



C. Protect vulnerable groups from the heat

Line of action: Improvement of thermal comfort in schools									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness			Calendar	Short term (2027) Medium term (2030) Long term (2035)		
						Duration	Continued Punctual		
Description and actions									
Adapting schools to climate change involves conducting the necessary assessments and implementing improvements to building envelopes and passive solutions (green roofs, natural ventilation protocols, etc.), installing climate control systems through the School Climate Plan program (a plan for public primary schools), and carrying out improvements in schoolyards (vegetation, shade, permeability, installation of water points, etc.). These actions must be accompanied by training and participation from the educational community. The actions planned in this area within the "Bressols pel Clima" strategy must also be implemented. Consider creating lines of aid also for secondary schools and for private and charter schools.									
Monitoring indicators	Number of schools adapted to climate change			Milestone					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		✓	Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)		✓			
	Lower energy or water consumption			Reduction of inequalities		✓	Job creation		✓
	Improvement of environmental quality		✓	Increased heat resilience		✓			
Operational information									
Target audience		Students and educational community							
Action proposed in the participatory process		Yeah							
Responsible area		Area Management of Culture, Education, Sports and Life Cycles; Management of Urban Services and Public Space Maintenance (Climate Change Office and Sustainability); Barcelona Education Consortium							
Other agents involved		Chief Architect's Office; Teaching Department (GENCAT)							



C. Protect vulnerable groups from the heat

Course of action		Improvement of thermal comfort in facilities intended for the elderly and people with disabilities					
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual		
Description and actions							
Develop a diagnosis of the current thermal comfort situation of residences that accommodate people in situations of vulnerability to heat (public and private residences, day centers, apartments, sheltered flats, etc.). Define a catalog of solutions with actions that will improve the thermal comfort of these spaces. Also assess the prioritization and implementation of shade structures. Explore the possibility of offering technical and financial assistance for the renovation of private residences to improve the thermal comfort of their residents. Working together with the other supralocal administrations (GENCAT) to design policies, incorporate measures into the reference regulations and facilitate the adaptation of these facilities to the heat.							
Monitoring indicators	Number of facilities where interventions have been carried out		Milestone				
Benefits		Environmental		Social		Economic	
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings	✔
			Improved well-being and health (mortality and morbidity)				
	Lower energy or water consumption		Reduction of inequalities		✔	Job creation	
	Improvement of environmental quality	✔	Increased heat resilience		✔		
Operational information							
Target audience		Users and staff of the residences					
Action proposed in the participatory process		Yeah					
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)					
Other agents involved		Management of Social Rights, Health, Cooperation and Community; Government of Catalonia					



C. Protect vulnerable groups from the heat

Course of action		Definition of a plan to improve the thermal comfort of municipal facilities intended for the care of vulnerable groups							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness			Calendar	Short term (2027) Medium term (2030) Long term (2035)		
						Duration	Continued Punctual		
Description and actions									
Assessment of the current situation and development of an action plan to ensure thermal comfort in facilities serving vulnerable groups: those providing services for social emergencies, children, the elderly, and Citizen Service Offices. Preparation of a catalog with proposed interventions.									
Monitoring indicators	Number of equipment seized Investment made in improving thermal comfort				Milestone				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption			Reduction of inequalities		✔	Job creation		
	Improvement of environmental quality			Increased heat resilience		✔			✔
Operational information									
Target audience		Vulnerable groups							
Action proposed in the participatory process		Yeah							
Responsible area		General Services Management							
Other agents involved		Urban Services and Public Space Maintenance Management (Barcelona Energy Agency); Districts; Area Management of Culture, Education, Sports and Life Cycles; Management of Social Rights, Health, Cooperation and Community							



C. Protect vulnerable groups from the heat

Action line: Training for caregivers							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation	Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness	Duration	Continued Punctual		
Description and actions							
Providing training on energy poverty and climate resilience for caregivers (e.g., home care services). The aim is to train these individuals and equip them with the tools to identify situations of energy poverty and provide them with information to help vulnerable people cope with the heat. This service complements other resources, such as energy advice centers and the network of climate shelters, which serve a particularly vulnerable segment of the population who, for various reasons, do not have access to them. This training can be extended to other services or facilities, such as senior centers.							
Monitoring indicators	Number of people trained			Milestone			
Benefits		Environmental		Social		Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility	✔	Cost savings	✔
				Improved well-being and health (mortality and morbidity)	✔		
	Lower energy or water consumption			Reduction of inequalities	✔	Job creation	
	Improvement of environmental quality			Increased heat resilience			
Operational information							
Target audience		Caregivers of vulnerable groups					
Action proposed in the participatory process		Yeah					
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)					
Other agents involved		Area Management of Social Rights, Health, Cooperation and Community					



C. Protect vulnerable groups from the heat

Action line: Strengthening of the PAE: energy rights against heat for all								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
					Duration	Continued Punctual		
Description and actions								
Specific actions to enhance community participation and improve thermal comfort in the homes of users of the PAE:								
- Promote the installation of thermal reduction elements in the homes of users of the PAE.								
Raise awareness through workshops and information campaigns on how to protect yourself from the heat at home with limited resources (close windows during peak hours, stay hydrated, use cross ventilation, etc.).								
- To promote the participation of PAE users in energy communities and shared use projects that involve their participation in the community and, at the same time, reduce the costs of refrigeration consumption in their homes.								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings		✔
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation		✔
			Increased heat resilience		✔			
	Improvement of environmental quality	✔						
Operational information								
Target audience		People at risk of energy poverty						
Action proposed in the participatory process		No						
Responsible area		Area Management of Social Rights, Health, Cooperation and Community						
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability, Barcelona Energy Agency)						



C. Protect vulnerable groups from the heat

Course of action		Other specific actions to protect people potentially vulnerable to heat						
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness		Duration	Continued Punctual		
Description and actions								
Strengthen and incorporate new actions and resources for the protection of people in vulnerable situations, seek synergies with existing social programs and projects, and assess the need to develop new projects with this objective: - Strengthen the Radar project during periods of extreme heat, intensifying calls and visits to people elderly people who live alone. - Strengthen the Telecare service by sending automated messages or calls with recommendations adapted to the health conditions of each user. - Update the censuses of people in vulnerable situations to prioritize their care during heat waves Identify people at risk. - To encourage and promote the creation of networks of neighborhood volunteers during the summer months who regularly visit people in vulnerable situations. - Provide specific aid to certain vulnerable groups for the installation of measures that help maintain a suitable temperature in homes. - Organize educational workshops on heat prevention (community training) with information sessions aimed at elderly people and caregivers. - Develop specific communication actions and resources for vulnerable groups (also in several languages), with specific heat recommendations, as well as the identification of the nearest climate shelters.								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect			Shared responsibility		✓	Cost savings	
				Improved well-being and health (mortality and morbidity)		✓		
	Lower energy or water consumption			Reduction of inequalities		✓	Job creation	
	Improvement of environmental quality			Increased heat resilience		✓		
Operational information								
Target audience		People potentially vulnerable to heat						
Action proposed in the participatory process		No						
Responsible area		Management of Social Rights, Health, Cooperation and Community; Management of Services Urban Planning and Public Space Maintenance (Climate Change and Sustainability Office)						
Other agents involved								



C. Protect vulnerable groups from the heat

Action line: Expansion of services on beaches for people with reduced mobility								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Improved knowledge and innovation Communication and awareness		Duration	Continued Punctual		
Description and actions								
Installation of a new assisted bathing point at Bogatell beach. Improvement of the bathing area at the Fòrum with the installation of an accessible changing room and improvement of the Barceloneta/Sant Miquel bathing area with a new accessible changing room. Development of a pilot footbridge for people with reduced mobility: definition of a footbridge solution suitable for Barcelona's beaches, designed according to the local coastal dynamics and the needs of potential user groups. Development of a prototype to test the solution (tender). Definition of the number of structures and priority locations, based on the available budget (ideally, one structure per beach).								
Monitoring indicators					Milestone			
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings	
				Improved well-being and health (mortality and morbidity)		✓		
	Lower energy or water consumption			Reduction of inequalities		✓	Job creation	
	Improvement of environmental quality			Increased heat resilience		✓		
Operational information								
Target audience		Vulnerable groups and people with reduced mobility						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Cycle of Water)						
Other agents involved		Management of Social Rights, Health, Cooperation and Community; Metropolitan Area of Barcelona						



C. Protect vulnerable groups from the heat

Line of action: Protection of homeless people									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
			Economic Improved knowledge and innovation		Duration	Continued Punctual			
Communication and awareness									
Description and actions									
Identify areas of the city that, due to their urban layout and the characteristics of the people who sleep there, are more susceptible to the effects of heat waves. In these areas, carry out a special prevention campaign and provide information on shelters, guidelines to be implemented, available equipment, how to respond to heat alerts, etc.									
To train and raise awareness among people working in municipal facilities that are climate shelters so that they can assist homeless people and encourage them to use the facilities, in order to avoid possible conflicts that may arise.									
To initiate a line of work between the Network for the Care of Homeless People (XAPSLL) and the Office of Climate Change and Sustainability of the Barcelona City Council to work collaboratively on the protection of this group in matters relating to heat.									
Monitoring indicators	Number of homeless people served Number of shelter workers trained		Milestone						
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower consumption energetic or of water			Reduction of inequalities			Job creation		
	Improvement of environmental quality			Increased heat resilience					
Operational information									
Target audience		Homeless people							
Action proposed in the participatory process		Yeah							
Responsible area		Area Management of Social Rights, Health, Cooperation and Community							
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)							



C. Protect vulnerable groups from the heat

Line of action: Adaptation of events organized abroad								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness		Duration	Continued Punctual		
Description and actions								
Establish an intersectoral working group to define recommendations that should be incorporated into the organization of outdoor events to ensure the health of workers, participants, and attendees, such as changes to schedules and times; create refreshment stations with multiple water fountains and shade, etc. Subsequently, in the medium term, an assessment will be made regarding the monitoring of its implementation, identifying proposals and improvement actions if necessary. Study the feasibility of having removable and reusable shade structures available to event organizers.								
Monitoring indicators	Number of events with heat adaptation measures		Milestone					
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption		Reduction of inequalities			Job creation		✔
	Improvement of environmental quality		Increased heat resilience		✔			
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		No						
Responsible area		Management of Social Rights, Health, Cooperation and Community (ASPB)						
Other agents involved		Area Management of Culture, Education, Sports and Life Cycles; Events Committee; Security, Prevention and Coexistence Management; Urban Services Management and Public Space Maintenance (Climate Change and Sustainability Office)						



C. Protect vulnerable groups from the heat

Line of action: Extension of opening hours for parks and gardens								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)			
				Duration	Continued Punctual			
Description and actions								
Modification of the opening hours of green spaces with city closure to allow their use during cooler hours, according to the priorities identified from the territory and adjusting them to the issues of staff conciliation. This action will be carried out in agreement with the affected districts to take into account all the information closest to the territory.								
Monitoring indicators	Number of parks with extended hours			Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)		✓			
	Lower energy or water consumption		Reduction of inequalities		✓	Job creation		
	Improvement of environmental quality		Increased heat resilience		✓			
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Parks and Gardens)						
Other agents involved		Districts						



D. Improve emergency response procedures and protocols in case of intense heat

Line of action: Improvement of protocols and contingencies for social care in emergency situations								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
	Communication and awareness		Duration	Continued Punctual				
Description and actions								
Coordination of the different departments of the Municipal Institute of Social Services to collect the various actions within the scope of action of the different emergency procedures in the face of extreme temperatures (Emergency Heat Plan), both existing and those not foreseen but necessary. Calling meetings, compiling proposals, collecting implementation indicators, providing communication tools to the population adapted to the different care profiles (elderly people, homeless people, etc.). Creation of a census of at-risk populations and unification of the various lists of vulnerable individuals to whom recommendations and advice are offered, in order to avoid duplication of messages. Initially, this will involve working with different municipal services, exploring the possibility of sharing user data with other administrations (e.g., the Department of Health of the Catalan Government). We urge the competent authorities to carry out risk assessments and contingency plans for facilities that house frail or heat-vulnerable people (e.g., residences).								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility		✓	Cost savings		
			Improved well-being and health (mortality and morbidity)		✓			
		Lower consumption energetic or of water		Reduction of inequalities		✓	Job creation	
				Increased heat resilience		✓		
Operational information								
Target audience		Citizenship						
Action proposed in the participatory process		No						
Responsible area		Area Management of Social Rights, Health, Cooperation and Community						
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); Government of Catalonia; Red Cross; Resources Management and Digital Transformation (Communication)						



D. Improve emergency response procedures and protocols in case of intense heat

Line of action: Improvement of the alert system for weather events									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
			Communication and awareness		Duration	Continued Punctual			
Description and actions									
Strengthen and improve the alert system by defining a strategy based on prediction, effective communication, and inter-institutional coordination. This strategy may include improvements in coordination among the different administrations involved, in the prediction and monitoring of extreme heat, and in public communication and awareness. Evaluate the possibility of implementing specific heat alerts based on health or incorporating variables such as humidity.									
Monitoring indicators				Milestone					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings			
			Improved well-being and health (mortality and morbidity)		✓				
		Lower energy or water consumption		Reduction of inequalities		✓	Job creation		
				Improvement of environmental quality		✓			
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		No							
Responsible area		Security, Prevention and Coexistence Management							
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); Area Management of Social Rights, Health, Cooperation and Community; Resource Management and Digital Transformation (Communication)							



D. Improve emergency response procedures and protocols in case of intense heat

Action line: Training of healthcare personnel								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness		Duration	Continued Punctual		
Description and actions								
Development of a training plan for healthcare personnel to prevent the health effects of heat. This plan should be structured and adapted to the needs of different healthcare professionals. Prior to its development, the participation of all stakeholders must be identified and organized.								
Monitoring indicators	Number of people trained			Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption		Reduction of inequalities			Job creation		✔
	Improvement of environmental quality		Increased heat resilience		✔			
Operational information								
Target audience		Healthcare personnel						
Action proposed in the participatory process		No						
Responsible area		Area Management of Social Rights, Health, Cooperation and Community						
Other agents involved		Barcelona Health Consortium, Urban Services and Maintenance Management Public Space (Climate Change and Sustainability Office)						



D. Improve emergency response procedures and protocols in case of intense heat

Course of action: Conducting a simulation exercise in Barcelona at 50 °C									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)				
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual				
Description and actions									
Conducting a drill to prepare for future extreme heat waves. Carrying out a large-scale crisis exercise in a specific area of the city, simulating a situation in which the temperature reaches 50°C. The objective will be to assess the city's and residents' capacity to respond to extreme weather conditions; anticipate potential challenges such as power outages, risks to essential service workers and municipal essential service contracts (cleaning, maintenance of public space, etc.), impacts on mobility, difficulties in transporting medicines, and the need to protect vulnerable people. Before conducting the on-site or desk-based drill, a simulation of a possible climate scenario will be carried out, along with the identification and confirmation of the sectors and stakeholders involved. Subsequently, working sessions will be organized with various stakeholders, as well as training and awareness workshops aimed at both the public and the stakeholders themselves. These stakeholders will include facilities serving vulnerable groups and educational centers, as well as critical urban service infrastructure. The simulation should identify weaknesses and develop strategies to improve the city's resilience to global warming, in order to protect the health and well-being of Barcelona's citizens in the future. In this regard, the results and conclusions of the simulation will be compiled into a final document with corresponding action proposals in those areas where potential improvements have been identified to mitigate the risks arising from high temperatures and their implementation (e.g., communication, information, data, coordination, technological tools, infrastructure, participation, etc.). In addition, the necessary resources, deadlines, and approximate budget for its implementation will be detailed.									
Monitoring indicators				Milestone	Conducting the drill				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower consumption energetic or water			Reduction of inequalities			Job creation		
	Improvement of environmental quality			Increased heat resilience					
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); Security, Prevention and Coexistence Management							
Other agents involved		Area Management of Social Rights, Health, Cooperation and Community; Management of Mobility, Infrastructure and Urban Services; Management of Commerce, Catering, Food and Markets; urban service companies (electricity, communications, etc.); private sector (tourism sector, local commerce, etc.); academic and research sector that contributes data, experience, studies and innovation							



Moving towards a new culture of heat through communication, training and participation

Line of action: Definition of an external communication plan

State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)
				Duration	Continued Punctual

Description and actions

Development of a communication plan to inform citizens about the Heat Plan and to carry out a sustained dissemination of the measures that will be included from now until 2030, framed within the Climate Plan strategy:

- Dissemination of the plan itself.
- Information and prevention advice, presented in an accessible format and in multiple languages: targeted campaigns for vulnerable groups and tourists; information leaflets with tips for preventing heatstroke and information on climate shelters. Distribution through districts, CUESB (Emergency Care Centers), social services centers, SAD (Home Help Service), Telecare, etc.
- Continuous updates to the barcelona.cat/refugisclimatics website. Updates to the shelter map on the website and in the La meva butxaca app.
- Communication in alert situations (improvements in the system used, development of applications and messaging services, etc.).
- Production of identification, support and merchandising items: refrigeration equipment, signage for shelter spaces, cups, fans, posters, etc. These are distributed in districts, CAP, OAC, PAE, Housing Offices, etc.
- Development of easy-to-use applications to locate available heat adaptation resources (nearest climate shelters, the coolest routes and itineraries to travel, the location of places where there are services such as drinking fountains, public toilets, etc.).
- Media and communication campaign: from June 15 to approximately September 15 (press, digital and advertising).
- Development of a new information campaign: pilot test of a campaign aimed at older people in coordination with Social Rights in CAP, senior centers or social services centers.

Monitoring indicators			Milestone			
		Environmental		Social		Economic
Benefits		Reduction of temperature and the urban heat island effect		Shared responsibility	✓	Cost savings
				Improved well-being and health (mortality and morbidity)	✓	
		Lower energy or water consumption		Reduction of inequalities	✓	Job creation
		Improvement of environmental quality		Increased heat resilience	✓	

Operational information

Target audience	Citizenship
Action proposed in the participatory process	Yeah
Responsible area	Resource Management and Digital Transformation (Communication)
Other agents involved	Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability; Parks and Gardens; Social Rights Area Management, Health, Cooperation and Community; Security, Prevention and Coexistence Management



Moving towards a new culture of heat through communication, training and participation

Line of action: Promotion of cultural change associated with heat							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness	Calendar	Short term (2027) Medium term (2030) Long term (2035)		
				Duration	Continued Punctual		
Description and actions							
Improve understanding of the risks of heat within the social fabric of the city (citizens, businesses, organizations, educational centers, etc.) through education and awareness programs. Incorporate the heat variable into the actions carried out within the framework of the Barcelona + Sustainable Program (e.g., in the project “Youth for the climate emergency”; advising entities, companies, educational centers, creation of large-format educational activities, programming of activities with environmental classrooms, writing of specific guides on how to combat heat in buildings with passive measures, offering of resources and incentives, climate subsidies, etc.). Develop specific training in urban resilience to recover a culture of heat resistance. Create a self-assessment checklist so that citizens can verify the resilience of their homes to heat. Disseminate case studies in which citizens and neighborhood associations have taken measures related to heat adaptation. Consider developing new resources to facilitate this change. Organize scientific conferences open to the public to share and improve knowledge and information on topics related to heat and possible adaptation actions in the city.							
Monitoring indicators	Number of impacts achieved		Milestone				
Benefits		Environmental			Social		Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings	✔
			Improved well-being and health (mortality and morbidity)		✔		
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation	✔
	Improvement of environmental quality	✔	Increased heat resilience		✔		
	Operational information						
Target audience		Citizenship					
Action proposed in the participatory process		Yeah					
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)					
Other agents involved		Area Management of Social Rights, Health, Cooperation and Community (ASPB); Management of Resources and Digital Transformation (Communication); other areas involved depending on the topic					



Moving towards a new culture of heat through communication, training and participation

Line of action: Development of citizen participation and experimentation actions									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
					Duration	Continued Punctual			
Description and actions									
Planning and organizing participatory activities, in collaboration with private sector stakeholders, neighborhood and social organizations, and research centers, to involve them in the design and implementation of heat adaptation solutions, improve knowledge, and obtain new information. Involving vulnerable communities to better address inequalities.									
Identification, through citizen marches or other participatory actions, of spaces where shade and other elements related to heat, such as cooling devices, can be located; carrying out citizen science studies on thermal comfort (outside or inside buildings), etc.									
Monitoring indicators	Number of actions performed			Milestone	At least one share per year				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption			Reduction of inequalities			Job creation		
				Improvement of environmental quality					
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		Management of Mobility, Infrastructure and Urban Services (Participation)							
Other agents involved		Reference services in the areas of action of the Heat Plan; entities and citizen organizations							



Promote research, knowledge and innovation on heat

Line of action: Updating of heat impact and vulnerability studies								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Communication and awareness		Duration	Continued Punctual		
Description and actions								
Update of climate projections in Barcelona according to the new scenarios of the Intergovernmental Panel on Climate Change (IPCC), with the collaboration of the Meteorological Service of Catalonia and other research centers. Update of studies on how rising temperatures will affect the city of Barcelona (in health, water supply security, sea level rise, flooding due to lack of drainage, biodiversity, fires, critical infrastructure, etc.), incorporating the perspective of intersectionality and determining how it affects different groups. With the collaboration of UN-Habitat, a cost-benefit analysis of the Climate Plan actions will be carried out, incorporating the heat variable. Update of the identification of priority risks in Barcelona, based on technical and scientific criteria, data and indicators on the expected impacts and vulnerability of the city, with a cost-benefit analysis that includes economic, environmental and social aspects, as well as action proposals and monitoring indicators. It also includes updating the heat vulnerability and risk maps. Zoning is being carried out.								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings	
				Improved well-being and health (mortality and morbidity)				
	Lower energy or water consumption			Reduction of inequalities			Job creation	
	Improvement of environmental quality			Increased heat resilience				
Operational information								
Target audience		Municipal and technical workers						
Action proposed in the participatory process								
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)						
Other agents involved		Barcelona Regional; Meteorological Service of Catalonia; Urban Services Management and Maintenance of Public Space (Barcelona Energy Agency, Barcelona Water Cycle, Parks and Gardens); Area Management of Social Rights, Health, Cooperation and Community; Management of Mobility, Infrastructure and Urban Services (Office of the Chief Engineer)						



Promote research, knowledge and innovation on heat

Course of action		Definition of a protocol for the system of monitoring the effects of climate change on health							
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness			Calendar	Short term (2027) Medium term (2030) Long term (2035)		
						Duration	Continued Punctual		
Description and actions									
Drafting the protocol for the climate change and health surveillance system, including piloting and evaluating its implementation. The surveillance system is being created to provide a tool for measuring, monitoring, and communicating the effects of climate change and its impact on health, using a social and climate justice methodology, to improve the health and well-being of the population of Barcelona. This surveillance system must be flexible and adaptable to changes in climatic and epidemiological patterns, with periodic reviews to improve its effectiveness. Evaluation of the effectiveness of heat emergency protocols.									
Monitoring indicators					Milestone				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings			
			Improved well-being and health (mortality and morbidity)		✓				
	Lower energy or water consumption		Reduction of inequalities		✓	Job creation			
	Improvement of environmental quality		Increased heat resilience		✓				
Operational information									
Target audience		Citizenship							
Action proposed in the participatory process		No							
Responsible area		Area Management of Social Rights, Health, Cooperation and Community (ASPB)							
Other agents involved									



Promote research, knowledge and innovation on heat

Line of action: Monitoring of heat-attributable mortality									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
					Duration	Continued Punctual			
Description and actions									
To better understand the impact of heat on health and how it evolves over the years, in order to analyze adaptation and the need to implement new measures, it is necessary to monitor temperature and burial records in the city, describe climate indicators annually, and estimate the risks of heat-related mortality and the impact of heat on mortality during the POCS activation period (May to September). Consider conducting a study on the impact of heat on health in indoor spaces, in collaboration with a research center.									
Monitoring indicators				Milestone					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings			
			Improved well-being and health (mortality and morbidity)		✓				
		Lower energy or water consumption		Reduction of inequalities		✓	Job creation		
				Increased heat resilience		✓			
Operational information									
Target audience		Technical staff, citizens							
Action proposed in the participatory process		No							
Responsible area		Area Management of Social Rights, Health, Cooperation and Community (ASPB)							
Other agents involved		Agents involved in obtaining data							



Promote research, knowledge and innovation on heat

Line of action: Definition of the thermal comfort monitoring system									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)				
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual				
Description and actions									
To define the most suitable temperature monitoring system for Barcelona, in order to better understand the city's situation regarding temperature, thermal comfort, and its vulnerability. This information will allow us to identify differences in behavior at the territorial level (neighborhoods and districts) and to monitor the effectiveness of implemented climate comfort improvement measures, the comfort provided by climate shelters, etc. This analysis should also allow the identification of existing systems and alternatives (sensors, technologies) and define the parameters to be measured (temperature, wind, humidity...) and the analysis tools, evaluation systems and monitoring to be implemented. Establish the protocol for collecting, processing and analyzing data that provides relevant information to guide public actions. The study will help define actions to be implemented within the framework of thermal comfort monitoring, such as improving the existing network of meteorological stations by incorporating additional citizen science stations. In this regard, a new station is planned for Mar Bella beach, and a buoy will also be deployed in the Port of Barcelona to measure water temperature.									
Monitoring indicators				Milestone					
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings			
			Improved well-being and health (mortality and morbidity)						
	Lower consumption energetic or water		Reduction of inequalities			Job creation			
	Improvement of environmental quality		Increased heat resilience						
Operational information									
Target audience		Municipal and technical workers, research centers, citizens							
Action proposed in the participatory process									
Responsible area		Urban Services and Public Space Maintenance Management (Evaluation and Management) Environmental, Office of Climate Change and Sustainability)							
Other agents involved		Area Management of Social Rights, Health, Cooperation and Community (ASPB); Barcelona Regional; Management of Urban Services and Maintenance of Public Space (Barcelona Water Cycle)							



Promote research, knowledge and innovation on heat

Line of action: Establishment of knowledge synergies between research centers								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation Communication and awareness		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
					Duration	Continued Punctual		
Description and actions								
Create a map of research centers, entities and universities working on projects in the field of heat adaptation. To co-produce a scientific agenda that bridges the gap between science and public action on heat-related issues, establishing research priorities that can be useful. To hold regular meetings to learn about current work, identify synergies, and share experiences, creating a network of knowledge and expertise in this field. Define a method to collect all available research on heat-related topics carried out by the Administration, research centers, universities, and experts, to avoid duplication and to share knowledge.								
Monitoring indicators				Milestone				
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect	✔	Shared responsibility		✔	Cost savings		✔
			Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption	✔	Reduction of inequalities		✔	Job creation		✔
			Increased heat resilience		✔			
	Operational information							
Target audience		Research centers, universities and public administration						
Action proposed in the participatory process		Yeah						
Responsible area		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability)						
Other agents involved		Research centers						



Promote research, knowledge and innovation on heat

Line of action: Launching urban challenges to find innovative solutions									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
			Improved knowledge and innovation Communication and awareness		Duration	Continued Punctual			
Description and actions									
Every two years, an urban challenge is launched to address the needs of the Winter Heating Plan. An urban challenge is defined as a complex urban problem or need that requires an innovative service, product, or methodology not yet available on the market. BIT Habitat promotes, designs, and adapts the format of urban challenges to meet the specific needs of each situation, and is committed to implementing pilot projects, analyzing the results, and measuring their impact.									
Monitoring indicators	Number of urban challenges completed and number of participating companies			Milestone	An urban challenge every two years				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		✔	Shared responsibility		✔	Cost savings		✔
				Improved well-being and health (mortality and morbidity)		✔			
	Lower energy or water consumption		✔	Reduction of inequalities		✔	Job creation		✔
				Improvement of environmental quality		✔			
	Operational information								
Target audience		Citizenship							
Action proposed in the participatory process		Yeah							
Responsible area		BIT HABIT							
Other agents involved		Urban Services and Public Space Maintenance Management; Architect's Management Head; Urban Planning and Housing Management; Social Rights and Health Area Management Cooperation and Community							



Promote research, knowledge and innovation on heat

Course of action	Feasibility study of different cold production and storage systems using renewable energy sources								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation		Calendar	Short term (2027) Medium term (2030) Long term (2035)			
			Communication and awareness		Duration	Continued Punctual			
Description and actions									
Feasibility study to carry out a pilot test to implement in a municipal facility a system for the production and accumulation of cold using different systems that utilize renewable generation sources, such as ice or salt accumulation tanks, adsorption pumps, etc.									
Monitoring indicators	Number of urban challenges completed and number of participating companies			Milestone	An urban challenge every two years				
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect		✔	Shared responsibility			Cost savings		✔
				Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption		✔	Reduction of inequalities			Job creation		
	Improvement of environmental quality		✔	Increased heat resilience		✔			
Operational information									
Target audience		Municipal buildings							
Action proposed in the participatory process		No							
Responsible area		Urban Services and Public Space Maintenance Management (Barcelona Energy Agency)							
Other agents involved		Any managing body of facilities where this pilot test can be implemented							



G. Adapt municipal service policies and management to the heat

Line of action: Improvement of thermal comfort in municipal buildings								
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic		Calendar	Short term (2027) Medium term (2030) Long term (2035)		
			Improved knowledge and innovation Communication and awareness		Duration	Continued Punctual		
Description and actions								
Development and implementation of criteria for improving thermal comfort within the framework of the new Energy Saving and Improvement Plan for Municipal Buildings (PEEMEM). This Plan will include measures to improve thermal comfort in different areas: insulation and energy efficiency, sustainable climate control systems (natural ventilation, use of renewable energy, efficient air conditioning, etc.), bioclimatic solutions and the use of vegetation (green roofs and facades, use of reflective materials, etc.), training and awareness-raising (improving the habits of municipal staff, promoting best practices through examples, etc.), and monitoring and evaluation of the implemented actions. The plan will include a diagnosis of the current situation and climate control problems in various types of municipal facilities, as well as a review of procedures to streamline incident management, including prioritization, scheduling, and budgeting for actions. Installation of shade structures in outdoor spaces (rooftops, patios, etc.) of facilities that house people sensitive or vulnerable to heat, as part of the Shade Plan. Beyond the fifteen school playgrounds already planned, work will be carried out in ten public facilities per year (daycare centers, residences for the elderly or people with disabilities, senior centers, assisted living facilities, etc.). The Shade Plan will also include measures to facilitate the installation of shade structures in private buildings.								
Monitoring indicators			Milestone					
Benefits		Environmental			Social			Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility			Cost savings		
			Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption		Reduction of inequalities			Job creation		
	Improvement of environmental quality		Increased heat resilience					
Operational information								
Target audience		Users of municipal buildings						
Action proposed in the participatory process		Yeah						
Responsible area		General Services Management; Urban Services and Maintenance Management Public Space (Barcelona Energy Agency)						
Other agents involved		Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); Urban Planning and Housing Management; Architect's Management Head; Area Management of Social Rights, Health, Cooperation and Community; Management of Culture, Education, Sports and Life Cycles; Barcelona Education Consortium; Government of Catalonia						



G. Adapt municipal service policies and management to the heat

Line of action: Definition of a protection plan for municipal staff									
State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic Improved knowledge and innovation	Calendar	Short term (2027) Medium term (2030) Long term (2035)				
			Communication and awareness	Duration	Continued Punctual				
Description and actions									
To coordinate and develop protection plans for personnel of municipal services and joint entities within the Occupational Risk Prevention Department of the Barcelona City Council, especially for those working outdoors (cleaning crews, maintenance of public spaces, gardening, construction, security, social emergencies, etc.). These plans must include, among other things, measures for adapting work schedules and organizing work, physical protection measures, equipment and clothing protection, health and safety protocols, training activities, etc., in accordance with the provisions of technical instruction IT4- PSSL-07 (Action in case of exposure to high temperatures and heat waves). In the event that a pregnant person or a person in the period of natural breastfeeding occupies the job position, the worker must communicate the situation of pregnancy or natural breastfeeding in accordance with the indications of instruction IT2-PSSL-12 available on the intranet. Ensure that companies providing services to the City Council that may have staff working outdoors incorporate heat-related health and safety criteria. If necessary, assist them in defining these measures (e.g., by drafting standard specifications).									
Monitoring indicators		Milestone							
Benefits		Environmental			Social			Economic	
	Reduction of temperature and the urban heat island effect			Shared responsibility			Cost savings		
				Improved well-being and health (mortality and morbidity)					
	Lower energy or water consumption			Reduction of inequalities			Job creation		
	Improvement of environmental quality			Increased heat resilience					
Operational information									
Target audience		Municipal staff / External staff (contracted companies)							
Action proposed in the participatory process		No							
Responsible area		People Management, Organization and Electronic Administration							
Other agents involved		Urban Services and Public Space Maintenance Management (Parks and Gardens; Cleaning and Waste Management, Urban Space); Area Management of Social Rights, Health, Cooperation and Community							



G. Adapt municipal service policies and management to the heat

Line of action: Definition and implementation of internal communication actions

State	Not started Started Consolidated	Type of measurement	Governance Regulatory or normative Operational and execution Economic	Calendar	Short term (2027) Medium term (2030) Long term (2035)
			Improved knowledge and innovation Communication and awareness	Duration	Continued Punctual

Description and actions

Develop diverse communication strategies to ensure that all municipal staff are informed about the risks of heat and the necessary prevention and protection measures, guaranteeing clear, effective communication tailored to the needs of different work groups. An annual schedule of these actions will be defined, utilizing existing channels (intranet, the "City Hall Information" mailbox, the monthly newsletter, etc.), as well as any other resources deemed appropriate and necessary.

Monitoring indicators

Milestone

Benefits		Environmental		Social		Economic	
	Reduction of temperature and the urban heat island effect		Shared responsibility		Cost savings		
	Lower consumption energetic or of water		Improved well-being and health (mortality and morbidity)				
			Reduction of inequalities		Job creation		
	Improvement of environmental quality		Increased heat resilience				

Operational information

Target audience	Personal municipal
Action proposed in the participatory process	
Responsible area	People Management, Organization and Electronic Administration
Other agents involved	Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); Area Management of Social Rights, Health, Cooperation and Community (ASPB)



G. Adapt municipal service policies and management to the heat

Line of action: Activation of economic and financing tools for heat adaptation						
State	Not started Started Consolidated	Type of measurement	Governance	Calendar	Short term (2027)	
			Regulatory or normative		Medium term (2030)	
			Operational and execution		Long term (2035)	
			Economic	Duration	Continued	
			Improved knowledge and innovation		Punctual	
			Communication and awareness			
Description and actions						
Mobilize economic resources to facilitate adaptation to heat in the domestic, community and business spheres, promoting co-responsibility, support for vulnerable groups and innovation in sustainability.						
The main actions that need to be implemented are:						
- Expansion of specific funds for climate adaptation improvements in buildings, especially in vulnerable areas (e.g., aid for efficient air conditioning, shading, natural ventilation, green roofs, etc.). - Tax breaks or incentives for companies or entities that implement heat resilience measures (vertical gardens, shaded spaces, adaptation of workstations, etc.). - Support for social and solidarity economy entities and projects that contribute to reducing climate vulnerability (energy cooperatives, care services adapted to the climate context, building rehabilitation, etc.). - Establishment of co-financing mechanisms with other administrations, access to European funds (NextGenerationEU, green funds, etc.) or public-private partnerships to deploy green infrastructures and promote climate adaptation projects or other community actions with high social and territorial impact. - Development of tools for the economic evaluation of the impacts of heat that allow guiding public investment decisions with criteria of climate justice and social return. - Promotion of green employment programs linked to climate adaptation: training and job placement in sectors such as urban gardening, installation of green roofs, maintenance of green infrastructure, efficient air conditioning, thermal insulation or management of reclaimed water.						
Monitoring indicators	Economic resources mobilized (euros)			Milestone		
Benefits		Environmental		Social		Economic
	Reduction of temperature and the urban heat island effect		Shared responsibility		Cost savings	
			Improved well-being and health (mortality and morbidity)			
	Lower energy or water consumption		Reduction of inequalities		Job creation	
	Improvement of environmental quality		Increased heat resilience			
Operational information						
Target audience	Entities, companies and public or private organizations, as well as municipal areas responsible for the implementation of heat adaptation actions					
Action proposed in the participatory process	No					
Responsible area	Economics and Economic Promotion Management					
Other agents involved	Urban Services and Public Space Maintenance Management (Change Office) Climate and Sustainability); other municipal areas					

Governance and monitoring of the Plan

To implement the Heat Plan, the necessary organizational changes and work methodologies must be incorporated to allow the heat variable to be included in municipal policies.

To this end, a monitoring committee will be established, comprised of members of the initial working group, which will meet regularly to promote and monitor the Plan. In addition, the creation of a climate control officer position with executive authority will be considered; this officer would be responsible for interdepartmental coordination with the relevant departments.

Mechanisms for effective coordination with other supra-local administrations will also be established, and active participation will take place in priority international city networks focused on heat-related issues and in any international cooperation projects that may be established. Participation in the C40 Cities Climate Leadership Group will be considered.

The capabilities of municipal technical staff to integrate the heat variable into their objectives and daily tasks will be improved, and the necessary tools will be provided to enable them to do so.

Finally, with the aim of continuous improvement, the implementation of the Plan will be monitored periodically. It will be updated based on the barriers and opportunities identified. The Plan will be fully reviewed every 5 years.

In addition to the specific indicators for monitoring the lines of action, priority will be given to monitoring the following indicators:

Table 4. Basic indicators for monitoring the Heat Plan.

Source: Own elaboration

Indicator Name	Definition	Periodicity Unit	
Context indicator			
Average annual temperature	Value obtained from the average of the average temperatures recorded each month of the year.	Annual	°C
Maximum annual temperature	The highest temperature recorded in a year.	Annual	°C
Number of hot and scorching days and number of tropical, scorching and hot nights	Number of days with temperatures exceeding 30°C (hot), 35°C (torrid) and nights with temperatures exceeding 20°C (tropical), 25°C (torrid) and 30°C (hot)	Annual	°C
Number and duration of heat waves	Three or more consecutive days with temperatures exceeding the 98th percentile of maximum temperatures.	Annual	Number
Effort indicator			
Number of activations of FEW	Number of activations and duration of emergency heat operational protocols	Annual	Number
Number of people or families in situations of vulnerability to heat contacted	According to the definition made in the emergency heat operation protocols.	Annual	Number
Number of facilities in vulnerable situations (schools and residences) adapted to climate change	Number of educational centers in which the playground has been transformed following the criteria of refuge schools, passive interventions have been carried out to improve thermal comfort or have participated in the School Climate Plan program. Residences that house elderly or disabled people and where interventions to improve thermal comfort have been carried out.	Annual	Number

Indicator Name	Definition	Periodicity Unit	
Number of shade elements installed	Number of shade structures installed in public spaces, children's play areas and educational centers, of a permanent, seasonal or ephemeral nature. Disaggregated by type. If available, shade area and energy production (kWh).	Annual	Number
Number of water games and cooling devices in the city	Number of water play areas for children broken down by districts.	Annual	Number
Impact indicator			
Heat-attributable mortality	Number of deaths according to the ASPB burial register: includes all people deceased who have been buried through the three main funeral homes of the city (Àltima, Mèmora and Interfuneràries).	Annual	Number
Degree of coverage of climate shelters	Percentage of population within a 5 and 10 minute walk of a climate shelter	Annual	%
Green space in the city	Green space in the total urban area and per inhabitant	Annual	m2 and m2 per inhabitant
Number of people trained on heat	Number of people trained on heat, whether they are part of the municipal staff, the Home Care Service, organizations or citizens.	Annual	Number

Overall budget

The Climate Plan government measure foresees an expenditure of **111,670,000 euros for the implementation of the Heat Plan.**

This value will be reviewed and updated as annual budgets are approved and the actions planned in this document are defined and implemented.

Calendar

	2027	2030	2035
a. Increase the heat resilience of public spaces and infrastructure and reduce the urban heat island effect			
1. Incorporation of heat adaptation criteria into urban development projects		X	
2. Definition of a plan to prevent damage to infrastructure from extreme heat situations X			
3. Reduction of direct heat emission sources			X
4. Increase in green space		X	
5. Improved adaptation of greenery to heat		X	
6. Implementation of the Plan for the recovery of green spaces affected by drought	X		
7. Improvement of thermal comfort on the coast and of coexistence during overnight stays	X		
8. Implementation of shadows		X	
9. Construction of permeable pavements, SUDS and reflectors		X	
10. Rationalization and optimization of ornamental fountains and improvement of public drinking water fountains	X		
11. Installation of refreshment devices and new water features	X		
12. Intensification of pest control in summer	X		
13. Odor control and reduction	X		
b. Improve thermal comfort in buildings			
1. Promotion of the rehabilitation and renovation of homes to adapt them to the heat	X		
2. Monitoring and development of new construction and rehabilitation regulations	X		
3. Promotion of reflective and cooling covers	X		
4. Development of a cooling plan for equipment used by citizens			X
c. Protect vulnerable groups from the heat			
1. Improvement and expansion of the climate shelter network	X		
2. Improvement of thermal comfort in schools			X
3. Improvement of thermal comfort in facilities intended for the elderly and people with disabilities		X	
4. Definition of a plan to improve the thermal comfort of municipal facilities intended for the care of vulnerable groups			X
5. Training for caregivers	X		
6. Strengthening the PAE: energy rights against heat for all	X		
7. Specific actions to protect people potentially vulnerable to heat		X	
8. Expansion of services on beaches for people with reduced mobility	X		
9. Protection of homeless people	X		
10. Adaptation of events organized abroad	X		
11. Extension of opening hours for parks and gardens	X		

	2027	2030	2035
d. Improve emergency response procedures and protocols in case of intense heat			
1. Improvements in internal protocols and contingencies for social care in emergency situations	X		
2. Improvement of the alert system for weather events	X		
3. Training of healthcare personnel	X		
4. Conducting a simulation exercise in Barcelona at 50 °C		X	
e. Move towards a new culture of heat through communication, training and participation actions			
1. Definition of an external communication plan		X	
2. Promoting cultural change related to heat		X	
3. Development of citizen participation and experimentation actions	X		
f. To promote research, knowledge and innovation on heat			
1. Update of heat impact and vulnerability studies	X		
2. Definition of a protocol for the system to monitor the effects of climate change on health	X		
3. Monitoring of heat-attributable mortality	X		
4. Improvement of thermal comfort monitoring systems	X		
5. Establishment of knowledge synergies between research centers	X		
6. Launching urban challenges to find innovative solutions	X		
7. Feasibility study of different cold production and storage systems using renewable sources	X		
g. Adapt municipal service policies and management to the heat			
1. Improvement of thermal comfort in municipal buildings		X	
2. Definition of a protection plan for municipal staff	X		
3. Definition and implementation of internal communication actions		X	
4. Activation of economic and financing tools for heat adaptation		X	



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Attachments

Annex I. Inspiring experiences from other cities

Melbourne

The Australian city of Melbourne has the "**Heat Health Plan for Victoria 2015**," a statewide heatwave action plan for community health service providers. The Victorian government website also offers several documents with protocols and information on how to prepare, geared towards different types of organizations and with specific guidelines for each type of organization.

The city's website features **a network of climate shelters** and tips for coping with the heat, as well as other tools related to managing heat waves.

It is worth noting that one of the documents (Health Preparedness Guidance Health and Community Services – November 2022) includes a list of people considered to be in a vulnerable situation, which has served as a source of inspiration for this Plan.

In Melbourne, green spaces have been a top priority as an adaptation and resilience strategy. Since 2017, they have had the "**Nature in the City Strategy**," which defines, through three objectives and six priorities, the strategy that will guide the planning, development, and management of the city's biodiversity and ecosystems over the next ten years. It also details a set of actions and targets to evaluate the success of its implementation. This plan outlines strategies to bring people closer to nature and make the city more resilient. They also have the "**Urban Forest Strategy**," which addresses the impact of drought and rising temperatures associated with the urban heat island effect.

More information:

[climate-change-mitigation-strategy-2050.pdf](#)

[Nature in the City Strategy](#)

[Urban Forest Strategy 2014](#)

Sydney

Sydney's Climate Action Plan, particularly the **Greater Sydney Heat Smart City Plan**, has been a source of inspiration for the development of Barcelona's Heat Plan. This plan, published in December 2024, is a multi-sectoral strategy that addresses extreme heat as one of the city's most serious natural hazards, with the aim of improving the metropolitan region's thermal resilience. Its development involved the collaboration of more than 40 organizations and 373 individuals, including leading figures in health, the environment, urban planning, business, government, and community.

Key actions include improving public spaces, creating green areas, and implementing heat-resistant infrastructure.

This collaborative and integrated methodology has been fundamental to Barcelona's Heat Plan, which aims to anticipate the effects of extreme heat through adaptive and mitigation measures. The creation of climate vulnerability maps, the identification of risk zones, and the implementation of nature-based solutions are some of the strategies adopted in Barcelona, inspired by the experiences and best practices of the Sydney Plan. Furthermore, the adaptation of international models reflects the importance of knowledge sharing and international cooperation in the fight against climate change.

More information: [WSROC Heat Smart City Plan_A4L_Digital_print friendly.pdf](#)

Medellín

Medellín's Climate Action Plan 2020-2050 is inspiring, as it combines a long-term vision with concrete actions that integrate climate justice, urban resilience, and citizen participation. The Plan focuses on three strategic pillars: mitigating greenhouse gas emissions, adapting to climate impacts, and inclusive action, ensuring that climate policies particularly benefit vulnerable communities. This structure allows for addressing the challenges of climate change in a comprehensive and equitable manner.

Furthermore, Medellín has developed tools such as the Climate Vulnerability Index and established clear goals for 2050, with intermediate targets for 2030. This methodology has also been implemented in Barcelona, particularly in identifying risk zones and implementing measures that are already effective, such as creating climate refuges and improving green infrastructure. The Medellín experience highlights the importance of long-term planning and coordination among different levels of government to address the effects of extreme heat and other impacts of climate change.

More information: [Medellín Climate Change Action Plan 2020-2050](#)

Boston

The **Heat Plan** developed in Boston in 2022 is a very detailed document that has also served as a model for the city of Barcelona. It includes a comprehensive analysis of heat in the city as a whole and by neighborhood, as well as of vulnerability to heat.

The fact that the Plan defines a series of benefits to identify how its strategies contribute to the city's objectives has been considered inspiring. Each strategy's fact sheet scores how well it contributes to each of the identified benefits. This idea has been adapted to Barcelona's Heat Management Plan.

We have analyzed the strategies defined in the Plan and compared them with our own, identifying those that can be part of the actions of the Heat Plan that we have already defined.

More information: [04212022_Boston Heat Resilience Plan_highres-with Appendix \(1\).pdf](#)

New York

The “**Cool Neighborhoods NYC**” plan includes actions such as intensive tree planting, the application of reflective paint to roofs (Cool Roofs), and the creation of green infrastructure to reduce the urban heat island effect. A vulnerability index is also used to prioritize interventions in the most at-risk neighborhoods.

Furthermore, the Plan incorporates social measures such as training for caregivers, the "Be a Buddy" program to promote neighborhood support, and improvements to cooling centers. The objectives are to protect public health, reduce inequalities, and increase urban resilience to climate change.

The actions that define their strategy for dealing with the heat are classified into those aimed at mitigating the heat, adapting to it, and monitoring it. Each of the strategies defined in the Plan has been analyzed and compared with those defined for Barcelona. In several cases, it has been possible to enrich the description of the actions defined in this Plan. Over the years, Barcelona has held several meetings with New York officials to exchange experiences in this area.

More information: [Cool_Neighborhoods_NYC_Report.pdf](#)

Paris

In Paris, everything related to heat is included in the **Climate Plan 2024-2030**, which defines the city's strategy for tackling climate change, from implementing shading systems to incentivizing all measures to reduce emissions. They have also developed a second document listing a total of 396 much more specific actions than those in the first document, all with the same objective of mitigating the effects of heat and protecting the city's residents.

In most cases, these are concrete actions that, more or less implicitly, were included in some of the lines of action designed for Barcelona, so the description of some of them has been enriched.

It is also worth noting that a 50°C Paris-style heatwave simulation was conducted in this city to prepare for potential extreme heat waves caused by climate change. This simulation served to test the response of public services and identify vulnerable areas. It also aimed to raise public awareness and promote urban adaptations. The objective was to anticipate an increasingly likely scenario. This simulation has served as a source of inspiration in Barcelona and has been used to define one of the actions described in this Plan. Communication has been established with key figures in the city of Paris to share knowledge for conducting the simulation in Barcelona, as well as for their Oasis project with schools and for issues related to shade.

More information: Paris [for the climate - City of Paris](#)

Vancouver

The Canadian city of Vancouver has a document outlining its strategy for mitigating and addressing climate change, as well as all related actions concerning heat. There is also a document with a highly innovative perspective that provides a brief overview of what the city has learned from past extreme heat events and describes the changes that need to be implemented in managing heat waves in the city in future situations. This document shares similarities with the action plan proposed by the Social Rights Department of the Barcelona City Council and will be used as a source of inspiration when it is updated.

More information: [Extreme heat | Vancouver Coastal Health](#)

<https://vancouver.ca/files/cov/2022-06-15-updates-to-extreme-heat-response-plans-for-2022.pdf>

Madrid

The Community of Madrid's **Action Plan for High Temperature Episodes** has the primary objective of reducing morbidity and, especially, mortality associated with heat waves. To this end, it establishes a monitoring and control system that allows for anticipating risk situations, coordinating interventions across different areas of the regional government, and providing preventative recommendations to the public and healthcare professionals. The Plan includes specific measures for social and healthcare centers, hospitals, schools, public transportation, and other sectors; it remains in effect until September 15. Its alert system has been analyzed and is considered a valuable reference document for future revisions.

More information: [plan-contra-calor.pdf](#)

London

The **London Climate Action Plan 2020-2027** comprehensively addresses adaptation to extreme heat in dense urban environments. It includes measures such as increasing urban green spaces, promoting green roofs and reflective surfaces, and implementing Sustainable Urban Drainage Systems (SUDS) to manage stormwater and reduce the urban heat island effect. This integrated methodology could be adapted to Barcelona to mitigate the effects of heat waves.

Furthermore, London has developed initiatives such as the Cool Spaces map to help residents find climate refuges during periods of extreme heat, and has conducted audits in care homes for the elderly to prevent overheating. Climate vulnerability maps have also been created to identify the most exposed areas and populations, enabling a more equitable and targeted response. These strategies are similar to those being developed in Barcelona to protect its most vulnerable communities from climate change and have been shared at several C40 meetings.

The document offers an innovative perspective comparing the risks and opportunities of climate change, something that has not been found in any other document.

More information: [Climate Action Strategy 2020-2027](#)

Athens

The case of Athens is of particular interest, as it is a city with very similar characteristics to Barcelona, especially regarding population density. Furthermore, Athens is a leading city in promoting the need to mitigate the effects of climate change and the necessity for cities to adapt to the new reality. It is one of the European cities that has suffered the most from extreme heat and has taken the most drastic measures to improve its resilience. Athens pioneered the creation of the position of Chief Heat Officer, with the aim of coordinating thermal resilience policies and raising public awareness of the risks of extreme heat. The Plan includes measures such as tree planting, the creation of green corridors, and the implementation of reflective roofing to reduce the urban heat island effect. Eleni Myrivili, the city's Chief Heat Officer and Director of Climate Affairs for the United Nations Human Settlements Programme, is a prominent advocate and a role model for her determination and values. Several meetings have been held with Athens through the C40 network of cities.

More information: [Urban Heat: The Athens Case Study](#)

Annex II. Compilation of specific actions of the Municipal Heat Wave Plan

Area of Social Rights, Health, Cooperation and Community September 2023



Ajuntament
de Barcelona

EMERGENCY OPERATIONS

OE 04

Heat warnings

Version No. 9

Delivery date: June 2023

PS/TIL/TS

Approved by the head of the DUES

1. JUSTIFICATION

Excessive and sustained heat leads to an increase in mortality and morbidity, especially in the **most vulnerable groups**, that is, people who, due to their socioeconomic situation or their health condition or age, suffer more directly from the effects of the heat.

It is expected that, due to the effects of the climate crisis, there will be an increasing frequency and intensity of episodes of intense heat during the daytime, and an increase in hot nights and scorching nights is also expected.

2. OBJECTIVE

The objectives of this protocol are:

- Minimize the negative effects of heat on the health of the population, especially in vulnerable groups, by providing intervention guidelines.
- Establish an intervention procedure regarding the measures to be applied in each phase activation due to heat warnings.
- Identify the most vulnerable people.

3. OBJECT

In a heat warning situation in the city, some people may be particularly affected.

The following are identified as **vulnerable groups** requiring social assistance:

- Elderly people living alone
- People with disabilities and limitations in mobility or self-care
- Babies and children
- People with chronic illnesses
- Mental health
- People experiencing homelessness

4. STRUCTURE AND ORGANIZATION OF THE PROTOCOL

Since this specific procedure is integrated into the Social Logistics Group's Action Plan, the structure and organization of the procedure are those indicated in the Plan itself.



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5. ACTIVATION CRITERIA

The following protocol activation criteria are established:

ACTIVATION PHASE	CRITERION
At the daytime level	
Preventive Phase: No danger. Heat warning for summer temperatures.	It is active from June 15th to September 15th
ALERT Phase: Warning for intense heat. High danger.	According to CECAT, it occurs when the maximum daily temperature is above the 98th percentile and below the 98th percentile + 2 °C of the maximum daily temperature of the summer months (in Barcelona, 98th percentile: 34 °C). At CUESB we will activate it when the maximum temperature reaches 34 °C, even if it is only for one hour of the corresponding time slot (from 6.00 to 24.00 h).
EMERGENCY Phase: Warning for very intense heat. Very high danger.	According to CECAT, the warning occurs when the temperature is above the 98th percentile of the daily maximum temperature + 2 °C (in Barcelona, 36 °C). At CUESB we will activate it when the maximum temperature reaches 36 °C, even if it is only for one hour of the corresponding time slot (from 6:00 to 24:00 h).
At night	
Warning for intense nighttime heat. High danger.	According to CECAT, it occurs when the minimum temperature exceeds the 98th percentile (in Barcelona, 26.5 °C) of the daily minimum temperature and is below the 98th percentile + 2 °C of the minimum. At CUESB we will activate it when the minimum temperature does not fall below 26 °C, during the corresponding night (from 24.00 to 6.00 h).
EMERGENCY Phase: Notice Due to very intense nighttime heat. Very high danger.	According to CECAT, the warning occurs when the minimum temperature exceeds the 98th percentile + 2 °C of the daily minimum temperature of the summer months (in Barcelona, 28.5 °C). At CUESB we will activate it when the minimum temperature does not fall below 28 °C, during the corresponding night (from 24.00 to 6.00 h).

Observation and recording schedule:

It should be noted that the highest temperature of the day or the corresponding time period must be considered, and this is the one that must be recorded, at each review time.



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08.00	11.30	19.00
Day, from 06:00 to 24:00 hours		
Night, from 24:00 to 06:00		

*According to the update on the Meteocat website.

IMPORTANT: Each phase will signify a new emergency number: preventive phase no. x, alert phase no. y, and emergency phase no. z, while the torrid night phase stands alone and could be due to high danger or very high danger, with a no. w.

When we activate the alert or emergency phases, if the temperatures include scorching nights, we will not activate it on its own, as it will be included in the text of the corresponding phase.

Once temperatures return to normal or to the previous phase, the current phase will be deactivated to return to the previous phase or to total deactivation.

IMPORTANT: The head of the Department of Social Emergencies and Urgent Care (DOS) may take into account other aspects, such as the social alarm generated, the existence of other simultaneous weather accidents, etc. when activating or deactivating this procedure.

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6. ACTIVATION PROCEDURES AND ACTIONS**PREVENTIVE PHASE****Safely**

It activates automatically on **June 15th and ends on September 15th: summer heat warning**. The following actions are implemented:

- Communication to the general population of the increase in temperatures and general recommendations through an information sheet available in all social services centers and municipal facilities for the elderly: day centers, geriatric residences, sheltered apartments, telecare and home help services, municipal social dining halls, etc.
- Communication to the general population of the Heat Plan, general recommendations and list of activated refuge spaces, through the ordinary municipal communication systems to the citizen: municipal websites, 010, etc.
- The permanent telephone service 900 70 30 30 (24 hours) is made available to the public
CUESB:
 - It will provide information and recommendations about the heat wave to people who request it.
 - It will provide information on available shelter spaces to those who request it.
 - If people with health problems who manifest physical discomfort are detected, coordination will be established with medical services.
 - Urgent social assessment will be activated for cases that present indicators of vulnerability.
- From the CUESB, you have to check the website of the Meteorological Service of Catalonia, Meteocat, daily to keep track of the weather forecast.

ALERT PHASE**High danger**

It is activated when observing, on the Meteocat website, the communication of a **meteorological situation in the high danger phase that may include a torrid night (point 5, Criteria for activating the schedule table)**.

The **head of the Emergency and Urgent Care Department (DUES)** and the municipal head on duty, as well as the CUESB Technical Coordination on duty, must be informed immediately .

- The head of the DUES, or the person to whom she delegates, must formally declare the activation.
- The head of the DUES, or the person to whom she delegates, must communicate the activation to the Civil Protection - Prevention Division of the SPEIS and to the other municipal operational groups and to the rest of the IMSS services.

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Keep in mind both the daytime and nighttime warnings.

The actions to be carried out during this phase **during the day** are as follows:

Preventive operation for vulnerable people

- The CUESB has designated the waiting room as a refuge space. Comfortable climate control and a water source must be provided. If necessary due to maximum occupancy, the space can be expanded by opening the dining room of the Short Stay Center (CEB).
In the context of a pandemic, it is necessary to guarantee the application of prevention and hygiene measures in public spaces.
- The head of the DUES, or the person to whom she delegates, can activate the collaboration agreement with the Red Cross in case of intense heat to assign volunteers who can provide support to the actions to assist the vulnerable population and the management of spaces.
- CUESB staff must respond to calls received for advice or support, guide people and, if necessary, travel to homes to make an assessment.
- Heat must be taken into account as one of the elements added to vulnerability in case of urgency or emergency, understanding that this factor, in some circumstances of fragility, can aggravate the situation of the person being cared for.
- In the case of individuals with home care services (SAD) who are identified as vulnerable, and for whom SAD professionals assess the need to provide a fan (either due to high temperatures in their homes or a lack of financial resources), one will be requested from the CUESB (Emergency Care Center). The CUESB must deliver it to their home to prevent them from having to travel during an alert phase.

In these cases, a home assessment by the CUESB will not be necessary, as it will be a case under follow-up and the Home Care Service will have already assessed the need. The CUESB will have to make a follow-up call after the ventilator delivery to verify that it has improved comfort at home.

From Monday to Friday, the Home Help Service (SAD) coordinator must submit this request via the 900 telephone number or the CUESB's generic email address. If they receive an urgent request from a Family Therapy Unit (TF) on a weekend or holiday, the CUESB will assess the situation at home and provide the ventilator, if necessary.

Specific operation on public roads

- The command of the operation must be located at the CUESB and the operation must be made up of the following people:
 - Head of the DUES, or the person to whom this person delegates, and municipal head on duty
 - CUESB Coordination or Deputy Technical Coordination



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- The head of the Emergency Medical Services (DUES), or their designee, may activate the collaboration agreement with the Red Cross in cases of intense heat to assign volunteers to the public outreach operation. Whether or not the agreement is activated, the Emergency Operations Center (CUESB) team must organize itself to carry out the outreach.
 - The public road information operation is established aimed at the synthetic people and vulnerable during the hottest part of the day in coordination with public space services.
 - The priority action areas are those indicated by the SASSEP team (service of intervention in public space) or others that are detected through reports from citizens or other operational groups.
- If in this operation any person is detected who presents physical discomfort and warns that their health may be at risk, 061 or 112 must be notified.
- When the emergency medical team (061) does not consider admitting the person being treated to a health center, but deems it appropriate to transfer them to an air-conditioned reception center, the person should be offered the option of moving to the CUESB.
- If any incident occurs during this operation related to the safety of social or public order emergency teams, the Urban Guard Command Center must be notified so that a patrol can be dispatched to the location of the incident.
- Daily, once the activity is completed, a report of the performance must be provided to the head of the DUES, the territorial heads and the management team of the CUESB (PROGESS).
- At the end of the state of alert, a single summary report of all actions taken will be issued, addressed to the people listed in the directory.

In case of a sweltering night, we add (from midnight to 6:00 a.m.):

- In the event of an alert from any citizen, if necessary, searches will be carried out in public areas and homes and, if necessary, an assessment of reception at the SEB.

EMERGENCY PHASE

Very high danger

It is activated when observing, on the Meteocat website, the communication of a **meteorological situation in the high danger phase that may include a torrid night (point 5, Criteria for activating the schedule table).**

The **head of the Emergency and Urgent Care Department (DUES)** and the municipal head on duty, as well as the CUESB Technical Coordination on duty, must be informed immediately .

- The head of the DUES, or the person to whom she delegates, must formally declare the activation.
- The head of the DUES, or the person to whom she delegates, must communicate the activation to the Civil Protection - Prevention Division of the SPEIS and to the other municipal operational groups and to the rest of the IMSS services.



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Regarding the actions to be carried out in this phase, those of the previous phase are maintained and expanded as follows:

- If necessary, and under prior coordination, the person is persuaded to move and is moved to the vulnerable people identified by social services who voluntarily accept this transfer to the CUESB refuge space or other activated spaces.
- CUESB can offer overnight stays after assessing the home, the public or in-person in cases where a social emergency is assessed and there is great fragility.
- Daily, once the activity is completed, a report of the performance must be provided to the head of the DUES, the territorial heads and the management team of the CUESB (PROGESS).
- At the end of the state of alert, a single summary report of all actions taken will be issued, addressed to the people listed in the directory.

In case of a sweltering night, we add (from midnight to 6:00 a.m.):

- In the event of an alert from any citizen, if necessary, searches will be carried out in public areas and homes and, if necessary, an assessment of reception at the SEB.

7. COMMUNICATION PROCEDURE

- In the event of a heat emergency being activated in the preventive, alert or emergency phase, an SMS and an email must be sent to all the people listed in the latest updated version of the CUESB directory.

SMS text:

Preventive phase:

Emergency activation. Level 1 for Emergency Manager, due to the preventive phase of the Heat Plan. Command: Monte Rovira.

Alert phase:

Emergency activation. Level 1 or 2 (depending on whether we activate the Red Cross), for the intense heat alert phase (daytime, day and scorching night, or scorching night) of the Heat Plan. Command: Monte Rovira.

Emergency phase:

Emergency activation. Level 1 or 2 (depending on whether we activate the Red Cross), for the emergency phase due to very intense heat (daytime, day and night scorching heat, or night scorching heat) of the Heat Plan. Command: Monte Rovira.



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The text of the email in case of **activation of the preventative phase** will be:

Good morning:

We inform you that, as of today, **June 15, 20xx**, following the Social Logistics Group's Action Plan, the **preventive phase** for summer temperatures is activated, which will remain in effect until September 15, 20xx.

The following actions are being implemented:

- a) Any technician from the municipal basic social care network who requires information on the effects of the heat wave and preventive measures can contact the CUESB.
- b) The general population must be informed about what the heat wave is and the general recommendations; for this purpose, we are sending you the attached sheet "Recommendations", as well as the list of refuge spaces available to the public. We will send brochures and posters to the municipal Social Services network.
- c) Each service of the municipal network of Social Services (CSS, SIS, SAIER, SARA, care for homeless people or those in situations of residential exclusion, in public spaces, etc.) must guarantee that it has identified the people or families who, given their personal, health, social, housing situation or lack thereof, dependency, etc., may become highly vulnerable to a heat wave.

If you have any questions or are in an emergency situation, you can contact CUESB by calling 900 703 030.

Thank you so much

The text in case of **activation of the alert phase** will be:

Good morning:

We inform you that, as of today, **xx of xx of 20xx**, following the Action Plan of the Social Logistics Group, the **alert phase** for high-risk heat warnings is activated .

Therefore, the following actions are implemented:

Preventive operation for vulnerable people

- IMSS (Mexican Social Security Institute) services should monitor individuals or families identified as vulnerable due to heat alerts. Where necessary, appropriate measures will be implemented to prevent them from leaving their homes: timely home catering, distribution of fans, personalized home care, etc.
- Specifically, the telecare services, through the switchboard, and the home help services, through the coordinators. Company technicians must establish an information and monitoring system for users.
- Attached is the list of activated shelter spaces available to the public.

Specific operation on public roads

- The public road information operation targeting the poor and vulnerable people on the fringe is established hot day in coordination with public space services.
 - The priority action areas are those indicated by the SASSEP team (public space intervention service) or others that are detected through reports from citizens or other operational groups.

If you have any questions or are in an emergency situation, you can contact CUESB by calling 900 703 030.

Thank you so much

Heat warnings

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Approved by the head of the DUES

The text in case of **activation of the emergency phase** will be:

Good morning:

Through this email we inform you that, as of today, **xx of xx of 20xx**, following the Action Plan of the Social Logistics Group, **the emergency phase is activated due to a heat warning with a very high danger.**

Therefore, the following actions are implemented:

- IMSS (Mexican Social Security Institute) services should maintain and expand monitoring of individuals or families identified as highly vulnerable due to heat alerts. Whenever possible, appropriate measures will be implemented to prevent them from leaving their homes: timely home catering, distribution of fans, personalized in-home care, etc.
- Vulnerable people, as determined by social services or emergency medical services, who voluntarily accept their transfer, are moved to the CUESB refuge space or other available spaces.
- If it is deemed necessary to extend the opening hours of the shelter spaces or open new ones, the procedure established in the work circuit with the Department of Urban Ecology must be followed (this point, depending on whether activation is deemed necessary, will be drafted according to the situation).
- The CUESB can expand the overnight accommodation offer in situations where it is assessed that there is a social emergency.

If you have any questions or are in an emergency situation, you can contact CUESB by calling 900 703 030.

Thank you so much

Annex III. Model of daily report of the Heat Plan, alert or emergency phase

Daily report of the Heat Plan: Emergency xx/23. Alert phase

Daily Heat Plan Report: Emergency xx/23. Emergency Phase

Fecha XX/XX/XXXX

1. People served in the CUESB, daytime from 6.00 to 22.00 h

People	Men	Women	Person non-binary	Total
People in the CUESB refuge space				
People receiving home care				
People served in public places, prospecting*				
Total number of people served by the CUESB				

* Does not include surveys if a street device is activated.

2. People attended at the CUESB, night shift from 10:00 p.m. to 6:00 a.m. (when the activation includes a night alert or emergency phase)

People	Men	Women	Person non-binary	Total
People in the CUESB refuge space				
People receiving home care				
People served in public places, prospecting				
Total number of people served by the CUESB				

3. Services activated

Type of services	Men	Women	Person non-binary	Total
Telephone information				
Accommodation at CUESB				
Accompanying people to safe spaces				
Fan delivery				
EMS activations				
Total resources activated				

4. People contacted on street devices (when activation includes street devices)

District	Men	Women	Person non-binary	Total
Old City				
Saints				
Saint Martin				
The Courts				
Sarrià-Sant Gervasi				
Saint Andrew				
New Neighborhoods				
The Eixample				
Grace				
Horta-Guinardó				
Total number of people				

4.1. Services activated on street devices

Type of services	Men	Women	Person non-binary	Total
Accommodation at CUESB				
Accompanying people to safe spaces				
EMS activations				
Total ACTIVATED RESOURCES				

5. Observations of the day

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Title

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